Unusual ordered phases of highly frustrated magnets: a review

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We review ground states and excitations of a quantum antiferromagnet on triangular and other frustrated lattices. We pay special attention to the combined effects of magnetic field h, spatial anisotropy R, and spin magnitude S. The focus of the review is on the novel collinear spin density wave and spin nematic states, which are characterized by *fully gapped* transverse spin excitations with $S^z = \pm 1$. We discuss extensively $R - h$ phase diagram of the antiferromagnet, both in the large-S semiclassical limit and the quantum $S = 1/2$ limit. When possible, we point out connections with experimental findings.

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I. INTRODUCTION

Frustrated quantum antiferromagnets have been at the center of intense experimental and theoretical investigations for many years. These relentless efforts have very recently resulted in a number of theoretical and experimental breakthroughs: quantum entanglement[1–3], density matrix renormalization group (DMRG) revolution and Z_2 liquids in kagome and $J_1 - J_2$ square lattice models[4–6], spin-liquid-like behavior in organic Mott insulators[7,8] and kagome lattice antiferromagnet herbertsmithite[9].

Along the way, a large number of frustrated insulating magnetic materials featuring rather unusual *ordered* phases, such as magnetization plateaux, longitudinal spin-density waves, and spin nematics, has been discovered and studied. It is these ordered, yet sufficiently unconventional, states of magnetic matter and theoretical models motivated by them that are the subject of this Key Issue article.

This review focuses on materials and models based on simple triangular lattice, which, despite many years of fruitful research, continue to supply us with novel quantum states and phenomena. Triangular lattice represents, perhaps, the most widely studied frustrated geometry[10–12]. Indeed, the Ising antiferromagnet on the triangular lattice was the first spin model found to possess a disordered ground state and extensive residual entropy[13] at zero temperature. While the classical Heisenberg model on the triangular lattice does order at $T = 0$ into a well-known 120° commensurate spiral pattern (also known as a three-sublattice or $\sqrt{3} \times \sqrt{3}$ state), the fate of the quantum spin-1/2 Heisenberg Hamiltonian has been the subject of a long and fruitful debate spanning over 30 years of research. Eventually it was firmly established that the quantum spin-$1/2$ model remains ordered in the classical 120° pattern[14–16]. Although the originally proposed resonating valence bond liquid[17,18] did not emerge as the ground state of the spin-1/2 Heisenberg model, such a phase was later found in a related quantum dimer model on the triangular lattice.[19]

It turns out that a simple generalization of the triangular lattice Heisenberg model whereby exchange interactions on the nearest-neighbor bonds of the triangular lattice take two different values – J on the horizontal bonds and J' on the diagonal bonds, as shown in Figure 1 – leads to a very rich and not yet fully understood phase diagram which sensitively depends on the magnitude of the site spin S and magnetic field h. Such a distorted, or spatially anisotropic, triangular lattice model interpolates between simple unfrustrated square lattice ($J = 0, J' \neq 0$), strongly frustrated triangular lattice

($J = J' \neq 0$) and decoupled spin chains ($J \neq 0, J' = 0$).

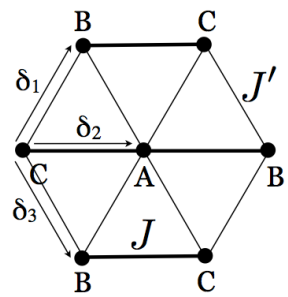

FIG. 1: Deformed triangular lattice. Solid (thin) lines denote bonds with exchange constant J (J') correspondingly. Also indicated are nearest-neighbor vectors δ_j as well as A, B and C sublattice structure.

A significant number of interesting experimental systems seems to fit into this simple distorted triangular lattice model with spin $S = 1/2$: Cs_2CuCl_4 (shows extended spinon continuum, $J'/J = 0.34$ is obtained from neutron scattering experiments in fully polarized state[20]), Cs_2CuBr_4 (shows magnetization plateau, $J'/J \approx 0.5$ based on considerations summarized in Section V A) and $Ba_3CoSb_2O_9$ (shows magnetization plateau, $J'/J \approx 1$[21]). Importantly, a number of very interesting organic Mott insulators of X[Pd(dmit)$_2$]$_2$ and κ-(ET)$_2$Z families[8,22] can also be approximately described by the spatially anisotropic $J - J'$ model with additional ring-exchange (involving four spins and higher order) terms. It is widely believed that these materials are weak Mott insulators in a sense of being close to a metal-insulator transition. As such, they are best described by distorted $t - t' - U$ Hubbard model, so that spatial anisotropy of exchanges follows from that of single particle hopping parameters, $J'/J \sim (t'/t)^2$. (See Section V C for the discussion of these parameters.)

The (intentionally) rather narrow focus of the review leaves out several important recent developments. Kagome lattice antiferromagnets represent perhaps the most notable omission. A lot is happening there, both in terms of experiments on materials such as herbertsmithite and volborthite[23] and in terms of theoretical developments[4,9]. This very important area of frustrated magnetism deserves its own review. We also have avoided another very significant area of development – systems with significant spin-orbit interactions. Progress in this area has been summarized in recent reviews[24,25].

The presentation is organized as follows: Section II contains brief review of the states of classical model in a magnetic field. Section III describes phase diagrams of the semi-classical $S \gg 1$ (Section III B) and $S = 1/2$ (Section III C) models. Novel ordered states, a collinear SDW and a spin nematic, which are characterized by the absence of $S = 1$ transverse spin excitations, are described in Section IV. Section V summarizes key experimental findings relevant to the review, including recent developments in organic Mott insulators.

II. CLASSICAL MODEL IN A MAGNETIC FIELD

Triangular antiferromagnets in an external magnetic field have been extensively studied for decades, and found to possess unusual magnetization physics that remains only partially understood. Underlying much of this interesting behavior is the discovery, made long ago,[26] that in a magnetic field, Heisenberg spins with isotropic exchange interactions exhibit a large *accidental* classical ground-state degeneracy. That is, at finite magnetic fields, there exists an infinite number of continuously deformable classical spin configurations that constitute minimum energy states, but are *not* symmetry related.

This degeneracy is understood by the observation that the Hamiltonian of the isotropic triangular lattice antiferromagnet in magnetic field $\mathbf{h}$ can be written, up to an unessential constant, as

$$H_0 = \frac{J}{2} \sum_{\mathbf{r}} \left[\mathbf{S_r} + \mathbf{S_{r+\delta_1}} + \mathbf{S_{r+\delta_2}} - \frac{\mathbf{h}}{3J} \right]^2. \qquad (1)$$

The sum is over all sites $\mathbf{r}$ of the lattice and nearest-neighbor vectors $\delta_{1,2}$ are indicated in the Figure 1. Note that Zeeman terms $\mathbf{S_r} \cdot \mathbf{h}$ appear three times for every spin in this sum, which explains the factor of $1/3$ in the $\mathbf{h}$ term in (1). We immediately observe that every spin configuration which nullifies every term in the sum (1) belongs to the lowest energy manifold of the model. Given the *side-sharing* property of the triangular lattice, so that fixing all spins in one elementary triangle fixes two spins in each of the adjacent triangles, sharing sides with the first one, this implies that all such states exhibit a three-sublattice structure and must satisfy

$$\mathbf{S}_A + \mathbf{S}_B + \mathbf{S}_C = \frac{\mathbf{h}}{3J}. \qquad (2)$$

This condition provides 3 equations for 6 angles needed to describe 3 classical unit vectors. In the absence of the field, the 3 undetermined angles can be thought of as Euler's angles of the plane in which the spins spontaneously form a three-sublattice 120° structure. However, this remarkable feature persists for $\mathbf{h} \neq 0$ as well. There, the symmetry of the Hamiltonian (1) is reduced to $U(1)$ but the degeneracy persists: one of the free angles can be thought as gauge degree of freedom to rotate all spins about the axis of the field $\mathbf{h}$, while the remaining two constitute the phenomenon of *accidental degeneracy*.

Remarkably, thermal (entropic) fluctuations lift this extensive degeneracy in favor of the two *coplanar* (Y and V states) and one *collinear* (UUD) spin configurations, shown in Figure 2. Symmetry-wise, coplanar states break two different symmetries – a discrete Z_3 symmetry, which corresponds to the choice of sublattice on which the down spin (in the case of Y state) or the minority spin (in the case of V state) is located, and a continuous $U(1)$ symmetry of spin rotations about the field axis. The collinear UUD state breaks only the discrete Z_3 symmetry (a choice of sublattice for the down spin). Selection of these simple states out of infinitely many configurations, which satisfy (2), by thermal fluctuations represents a textbook example of the 'order-by-disorder' phenomenon[27].

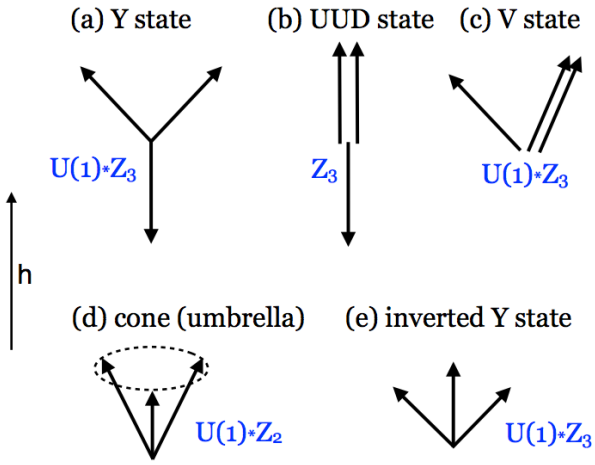

FIG. 2: Various spin configurations from the classical ground state manifold. (a) coplanar Y state, (b) collinear UUD (up-up-down), (c) coplanar V state, (d) non-coplanar cone (umbrella) state, (e) coplanar inverted Y state. Also indicated (in blue) are the order parameter manifolds for each of the states. Magnetic field h is directed vertically (along z axis).

The resulting enigmatic phase diagram, first sketched by Kawamura and Miyashita in 1985, Ref. 26, continues to attract much attention - and in fact remains not fully understood. Figure 3 shows the result of recent simulations[28], which studied critical properties of various phase transitions in great detail. Th obtained phase diagram differs in one important aspect from the original suggestion – it is established now that there is no direct transition between the Y state and the paramagnetic phase (very similar results were obtained in an extensive study[29]). The two phases are separated by the intervening UUD state which extends down to lowest accessible field values and before the transition to the paramagnetic state. Figure 3 also shows that of all entropically selected state, the UUD state is most stable - it extends to higher temperature T than either Y or V states.

The UUD is also the most 'visible' of the three selected states - it shows up as a plateau-like feature in the magnetization curve $M(h)$, see Figure 4 which shows experimental data for a $S = 5/2$ triangular lattice antiferromagnet $RbFe(MoO_4)_2$[30]. Notice that a strict magnetization plateau at $1/3$ of the full (saturation) value $M_{\rm sat}$, $M = M_{\rm sat}/3$, is possible only in the quantum problem (*i.e.* the problem with finite spin S), when all spin-changing excitations with $S^z \neq 0$ are characterized by *gapped* spectra, and at absolute zero $T = 0$, when no excitations are present in the ground state (see next Section III for complete discussion). At any finite T thermally excited spin waves are present and lead to a finite, albeit different from the neighboring non-plateau states, slope of the magnetization $M(h)$. In the case of the classical problem we review here the gap in spin excitation spectra is itself T-dependent and disappears as $T \to 0$: as a result its magnetization 'quasi-plateau' too disappears in the $T \to 0$ limit. However, at any finite T, less than that of the transition to the paramagnetic state, the slope of the $M(h)$ for the UUD state is different from that of the Y and V states[31].

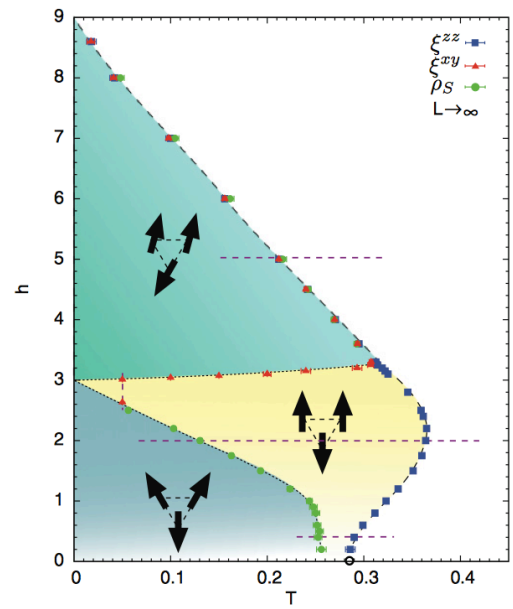

FIG. 3: Magnetic field phase diagram of the classical triangular lattice antiferromagnet. [Adapted from Seabra *et al.*, Phys. Rev. B **84**, 214418 (2011). Copyright 2011 by the American Physical Society.] Transition points determined by the Monte Carlo simulations are shown by filled symbols. Continuous phase transitions are drawn with a dashed line, while Berezinskii-Kosterlitz-Thouless phase transitions are drawn with a dotted line. For fields $h \leq 3$ a double transition is found upon cooling from the paramagnet, while for $h \geq 3$ only a single transition is found. Behavior of the phase transition lines in the low-field region $h \leq 0.2$, which is left unshaded in the diagram, is not settled at the present. See Ref.32 for the recent study of $h = 0$ line.

A. Spin chirality and emergent Ising orders

Understanding the symmetries broken by a particular ordered state amounts to specifying an *order parameter manifold* associated with this state. Thus, order parameter manifold of the 3-fold degenerate UUD state is Z_3, while that for the coplanar Y, V and inverted Y states in Figure 2 is $U(1) \times Z_3$. The non-coplanar cone state is characterized by $U(1) \times Z_2$ manifold where Z_2 is associated with spontaneous selection of the *vector chirality* $\boldsymbol{\kappa}_{\mathbf{rr'}} = \mathbf{S_r} \times \mathbf{S_{r'}}$, which is the sense in which spins rotate - clockwise or counterclockwise - as one moves between the sites of the elementary triangle of the lattice. Closely related to it is the *scalar chirality*, $\chi = \mathbf{S_r} \cdot \mathbf{S_{r+\delta_1}} \times \mathbf{S_{r+\delta_2}}$, which measures the volume enclosed by the three spins of the elementary triangle and, similarly to $\boldsymbol{\kappa}$, is sensitive to the sense of spin's rotation in the $x-y$ plane.

Composite nature of the order parameter manifold opens up a possibility of unusual sequence of phase transitions between the high temperature paramagnetic phase and the low temperature magnetically ordered phase. Namely, in the case of the non-coplanar cone state one can imagine existence of the in-

termediate chiral state with finite spin chirality $\langle \boldsymbol{\kappa} \rangle \neq 0$ but no long-ranged magnetic order $\langle \mathbf{S}_{\mathbf{r}} \rangle = 0$.

Reference 33 was perhaps one of the first studies of this possibility. Interestingly, the system studied there was antiferromagnetic XY model of spins on triangular lattice. Its magnetic field h - temperature T phase diagram is composed of the same phases, Y, UUD and V, that show up in the phase diagram of the Heisenberg model in the magnetic field, Figure 3. (The counting of degeneracies is of course different.) Extensive later studies, which are reviewed in[34], have indeed identified classical chiral spin liquid state in a narrow temperature interval $T_{\rm KT} < T < T_{\rm Ising}$: here $T_{\rm Ising}$ denotes the onset of the chiral long-range order, $\langle \boldsymbol{\kappa} \rangle \neq 0$, while the spin (quasi) long-range order emerges only below smaller $T_{\rm KT}$.[35,36]

Ising transitions associated with composite order parameters are also present in Heisenberg spin models and are often identified with emerging *nematic* orders (reviewed in more details in Section IV B)[37–40]. Notice that since Heisenberg spins do not order in two dimensions at any finite temperature, the Ising transition of the emergent composite order parameter is the only thermodynamic phase transition present in the system.

Regarding the present case of the classical Heisenberg model on the two-dimensional triangular lattice, Kawamura and Miyashita[41] have noticed long ago that the order parameter manifold of the model is isomorphic to the three-dimensional rotation group $SO(3)$ (rotations of a rigid object) and admits stable point defects. In fact, these defects are Z_2 vortices of the spin chirality. Correspondingly, they have identified a finite temperature peak in the specific heat, at $T_v \approx 0.285J$ and $h = 0$, observed in their Monte Carlo simulations, with the Kosterlitz-Thouless Z_2-vortex unbinding transition.[41] It is now established that unlike the case of the usual Kosterlitz-Thouless vortex-unbinding transition[34], here the spin correlation length remains finite (albeit very large) below T_v, resulting in a disordered "spin-gel" state[32]. Whether the change from high-temperature vortex-dominated regime to the low-temperature spin-fluctuation-dominated one is a true transition or a sharp crossover is perhaps the most outstanding and still not resolved issue[42–46], associated with the phase diagram presented in Figure 3. Experimental ramifications of different scenarios are reviewed in[47].

B. Spatially anisotropic triangular lattice antiferromagnet

A classical system with spatially anisotropic interactions offers an interesting generalization of the 'order-by-disorder' phenomenon. Consider slightly deformed triangular lattice, with $J' < J$ (see Fig. 1). An arbitrary weak deformation lifts, at $T = 0$, the accidental degeneracy in favor of the simple non-coplanar umbrella state (configuration 'd' in Fig. 2) in the whole range of h below the saturation field. The energy gain of this well-known spin configuration is of the order $(J - J')^2/J$. One thus can expect that, for sufficiently small difference $R = J - J'$, entropic fluctuations, which favor coplanar and collinear spin states at a finite T, can still overcome this classical energy gain and stabilize collinear and coplanar states *above* some critical temperature which can be estimated as $T_{\rm umb-uud} \sim R^2/J$. As a result, the UUD phase 'decouples' from the $T = 0$ axis. The leftmost point of the UUD phase in the $h - T$ phase diagram now occurs at a finite $T_{\rm umb-uud}$ as was indeed observed in Monte Carlo simulation of the simple triangular lattice model in Ref.31 (see Fig.10 of that reference), as well as in a more complicated ones, for models defined on deformed pyrochlore[48] and Shastry-Sutherland[49] lattices. It is worth noting that the roots of this behavior can be traced to the famous Pomeranchuk effect in ^{3}He, where the crystal phase has higher entropy than the normal Fermi-liquid phase. As a result, upon heating, the liquid phase *freezes* into a solid[50,51]. In the present classical spin problem it is 'superfluid' umbrella phase which freezes into a 'solid' UUD phase upon heating[31] (see discussion of the relevant terminology at the end of Section III A below).

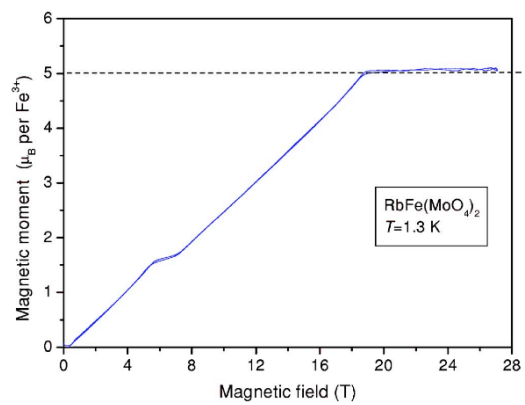

FIG. 4: Magnetization curve $M(h)$ of a spin-5/2 antiferromagnet $RbFe(MoO_4)_2$ at $T = 1.3$K. [Adapted from Smirnov *et al.*, Phys. Rev. B **75**, 134412 (2007). Copyright 2007 by the American Physical Society.]

III. QUANTUM MODEL IN MAGNETIC FIELD

A. Isotropic triangular lattice

Much of the intuition about selection of coplanar and collinear spin states by thermal fluctuations applies to the most interesting case of *quantum* spin model on a (deformed) triangular lattice. Only now it is quantum fluctuations (zero-point motion) which differentiate between different classically-degenerate (or, nearly degenerate) states and lift the accidental degeneracy.

One of the first explicit calculations of this effect was carried out by Chubukov and Golosov[52], who used semiclassical large-S spin wave expansion in order to systematically separate classical and quantum effects. This well-known technique relies on Holstein-Primakoff representation of spin operators in terms of bosons. The representation is nonlinear, $S^z_{\mathbf{r}} = S - a^\dagger_{\mathbf{r}} a_{\mathbf{r}}$, $S^+_{\mathbf{r}} = (2S - a^\dagger_{\mathbf{r}} a_{\mathbf{r}})^{1/2} a_{\mathbf{r}}$, and leads, upon expansion of square roots in powers of small parameter $1/S$, to bosonic spin-wave Hamiltonian $H = E_{\rm cl} + \sum_{k=2}^{\infty} H^{(k)}$. Here

$E_{\rm cl}$ stands for the classical energy of spin configuration, which scales as S^2, while each of the subsequent terms $H^{(k)}$ are of k-th order in operators $a_{\mathbf{r}}$ and scale as $S^{2-k/2}$. Diagonalization of quadratic term $H^{(2)}$ provides one with the dispersion $\omega_{\mathbf{k}}^{(m)}$ of spin wave excitations ($\mathbf{k}$ is the wave vector and m is the band index), in terms of which quantum zero-point energy is given by $\langle H^{(2)}\rangle = (1/2)\sum_{m,\mathbf{k}}\omega_{\mathbf{k}}^{(m)}$. This energy scales as S, and thus provides the leading quantum correction to the classical ($\propto S^2$) result.

The main outcome of the calculation[52] is the finding that quantum fluctuations also selects coplanar Y and V and collinear UUD states (states a, b and c in Figure 2) out of many other classically degenerate ones. The authors also recognized the key feature of the UUD state - being collinear, this state preserves $U(1)$ symmetry of rotations about the magnetic field axis. The absence of broken continuous symmetry implies that spin excitations have a finite energy gap in the dispersion. This expectation is fully confirmed by the explicit calculation[52] which finds that the gaps of the two low energy modes are given by $|h - h^0_{c1,c2}|$, where the lower/upper critical fields are given by $h^0_{c1} = 3JS - 0.5JS/(2S)$ and $h^0_{c2} = 3JS + 1.3JS/(2S)$, correspondingly. (The third mode describes a high-energy precession with energy hS.) The uniform magnetization M, being the integral of motion, remains at $1/3$ of the maximum (saturation) value, $M = M_{\rm sat}/3$, in the UUD stability interval $h^0_{c1} < h < h^0_{c2}$.

As a result, magnetization curve $M(h)$ of the quantum triangular lattice antiferromagnet is non-monotonic and exhibit striking 1/3 *magnetization plateau* in the finite field interval $\Delta h = h^0_{c2} - h^0_{c1} = 1.8JS/(2S)$. It is worth noting that this is not a narrow interval at all, $\Delta h/h_{\rm sat} = 0.2/(2S)$ in terms of the saturation field $h_{\rm sat} = 9JS$. Extending this large-S result all way to the $S = 1/2$ implies that the magnetization plateau takes up 20% of the whole magnetic field interval $0 < h < h_{\rm sat}$.

Numerical studies of the plateau focus mostly on the quantum spin-1/2 problem (numerical studies of higher spins are much more 'expensive') and confirm the scenario outlined above. The plateau at $M = M_{\rm sat}/3$ is indeed found, and moreover its width in magnetic field agrees very well with the described above large-S result, extrapolated to $S = 1/2$.[53–58]

The pattern of symmetry breaking by the coplanar/collinear states is described by the following spin expectation values

$$\text{Y state,}\quad 0 < h < h^0_{c1} : \langle S^+_{\mathbf{r}}\rangle = ae^{i\varphi}\sin[\mathbf{Q}\cdot\mathbf{r}],$$
$$\langle S^z_{\mathbf{r}}\rangle = b - c\cos^2[\mathbf{Q}\cdot\mathbf{r}]; \quad (3)$$
$$\text{UUD state,}\quad h^0_{c1} < h < h^0_{c2} : \langle S^+_{\mathbf{r}}\rangle = 0,$$
$$\langle S^z_{\mathbf{r}}\rangle = M - c\cos[\mathbf{Q}\cdot\mathbf{r}]; \quad (4)$$
$$\text{V state,}\quad h^0_{c1} < h < h_{\rm sat} : \langle S^+_{\mathbf{r}}\rangle = ae^{i\varphi}\cos[\mathbf{Q}\cdot\mathbf{r}],$$
$$\langle S^z_{\mathbf{r}}\rangle = b - c\cos^2[\mathbf{Q}\cdot\mathbf{r}]. \quad (5)$$

Here the ordering wave vector $\mathbf{Q} = (4\pi/3, 0)$ is commensurate with the lattice which results in only three possible values that the product $\mathbf{Q}\cdot\mathbf{r} = 2\pi\nu/3$ can take ($\nu = 0, 1, 2$), modulo 2π. Angle φ specifies orientation of the ordering plane for coplanar spin configurations within the $x - y$ plane, M is magnetization per site, and parameters a, b, c are constants dependent upon the field magnitude.

Connection with 'super' phases of bosons: Eq.(4) identifies UUD state as a collinear ordered state which can be thought of as *solid*. Its 'density' $\langle S^z_{\mathbf{r}}\rangle$ is *modulated* periodically, relative to the uniform value M, as $\cos[\mathbf{Q}\cdot\mathbf{r}]$, as is appropriate for the solid, and as a result its local magnetization follows simple 'up-up-down' pattern within each elementary triangle. This state respects $U(1)$ symmetry of rotations about S^z axis, as follows from the very fact that it has no order in the transverse directions, $\langle S^+_{\mathbf{r}}\rangle = 0$.

The coplanar Y and V states break this $U(1)$ symmetry by spontaneously selecting ordering direction in the spin $x - y$ plane, *i.e.* by selecting angle φ in (3),(5). Note that at the same time they are characterized by the modulated density $\langle S^z_{\mathbf{r}}\rangle$, which makes them *supersolids*: the superfluid order (magnetic order in the $x - y$ plane as selected by φ) co-exists with the solid one (modulated z component, or density).

This useful connection is easiest to make precise[59] in the case of $S = 1/2$ when the following mapping between a *hard-core* lattice Bose gas an a spin-1/2 quantum magnet can easily be established:

$$a^+_{\mathbf{r}} \leftrightarrow S^+_{\mathbf{r}}, a_{\mathbf{r}} \leftrightarrow S^-_{\mathbf{r}}, n_{\mathbf{r}} = a^+_{\mathbf{r}}a_{\mathbf{r}} \leftrightarrow S^-_{\mathbf{r}} - 1/2. \quad (6)$$

The superfluid order is associated with finite $\langle a^+_{\mathbf{r}}\rangle$ while the solid one with modulated (with momentum $2\mathbf{Q}$ in our notations) boson density $\langle n_{\mathbf{r}}\rangle$.

B. Spatially anisotropic triangular antiferromagnet with $J' \neq J$

Consider now a simple deformation of the triangular lattice which makes exchange interaction on diagonal bonds, J', different from those on horizontal ones J, so that $R = J - J' \neq 0$. This simple generalization of the Heisenberg model leads to surprisingly complicated and not yet fully understood phase diagram in the magnetic field (h) - deformation (R) plane.

Semiclassical ($S \gg 1$) analysis of this problem is complicated by the fact that arbitrary small $R \neq 0$ removes accidental degeneracy of the problem in favor of the unique non-coplanar and incommensurate cone (umbrella) state, state 'd' in Fig. 2, which was discussed previously in Sec. II. This simple state gains energy of the order $\delta E_{\rm class} \sim S^2R^2/J$ per spin. Its structure is described by

$$\langle \mathbf{S}_{\mathbf{r}}\rangle = M\hat{z} + c(\cos[\mathbf{Q}'\cdot\mathbf{r} + \varphi]\hat{x} + \sin[\mathbf{Q}'\cdot\mathbf{r} + \varphi]\hat{y}), \quad (7)$$

where classically the ordering wave vector $\mathbf{Q}' = (2\cos^{-1}[-J'/2J], 0)$ is a continuous function of J'/J. Spontaneous selection of the ordering phase φ makes the cone state a spin *superfluid* (note that here the density $\langle S^z_{\mathbf{r}}\rangle = M$ is constant). In addition, this non-coplanar state is characterized by the finite scalar chirality, $\chi \sim \mathbf{S}_{\mathbf{r}}\cdot\mathbf{S}_{\mathbf{r}+\delta_1}\times\mathbf{S}_{\mathbf{r}+\delta_2} \sim M\sin[Q'/2]$. Observe that under $\mathbf{Q}' \to -\mathbf{Q}'$ replacement chirality χ changes sign, while the ground state energy does not change. Hence, together with breaking spin-rotational $U(1)$ symmetry, the cone state also spontaneously breaks Z_2 symmetry (spatial inversion).

This consideration allows us to identify the *order parameter manifold* for this state as $U(1) \times Z_2$.

At the same time, for sufficiently small R, the quantum energy gain due to zero-point motion of spins, which is of the order $\delta E_{\rm q} \sim SJ$ per spin, should be able to overcome $\delta E_{\rm class}$ and still stabilize one of the coplanar/collinear states considered in Sec. III A above. Comparing the two contributions, we conclude, following Ref. 60, that the classical-quantum competition can be parameterized by the dimensionless parameter $\delta \sim \delta E_{\rm class}/\delta E_{\rm q} \sim S(R/J)^2$. (In the following, we will use more precise value $\delta = (40/3)S(R/J)^2$, with numerical factor $40/3$ as introduced in[60] for technical convenience.)

Explicit consideration of this competition is rather difficult due to complicated dependence of the parameters of the coplanar states on magnetic field h and exchange deformation R. However, inside the $M = M_{\rm sat}/3$ magnetization plateau phase, the spin structure is actually pretty simple, as equation (4) shows, which suggests that UUD state can be used as a convenient starting point for accessing more complicated states. This approach, which amounts to the investigation of the *local stability* of the UUD phase, was carried out in[60]. Similar to[52], the calculation is based on the three-spin UUD unit cell, resulting in three spin wave branches. One of these, describing total spin precession, is a high-energy mode not essential for our analysis, while the two others, which describe relative fluctuations of spins, are the relevant low-energy modes. They will be called mode 1 and mode 2 below (creation operators $d^\dagger_{1,\mathbf{k}}$ and $d^\dagger_{2,\mathbf{k}}$, correspondingly).

Within the UUD plateau phase, which is bounded by FACBG lines in Figure 5, both low energy spin wave modes remain gapped. As discussed above, the only symmetry this collinear state breaks is Z_3. The gaps of the low-energy spin wave modes are given by $|h - h_{c1,c2}|$, where now the lower and upper critical fields $h_{c1,c2}(\delta)$ depend on the dimensionless deformation parameter δ. The closing of the gap at the lower h_{c1} (upper h_{c2}) critical fields of the plateau implies Bose-Einstein condensation (BEC) of the appropriate magnon mode (mode 1 or mode 2, correspondingly) and the appearance of the transverse to the field spin component $\langle S^+_{\mathbf{r}} \rangle \neq 0$, see (3) and (5). This BEC transition breaks spin rotational $U(1)$ symmetry via spontaneous selection of the 'superfluid' phase φ. The spin structure of the resulting 'condensed' state sensitively depends on the wave vector $\mathbf{k}_1(\mathbf{k}_2)$ of the condensed magnon.

Key results of the large-S calculation in Refs. 60–62 can now be summarized as follows:

(**a**) In the interval $0 < \delta \leq 1$ the lower critical field is actually independent of δ, $h_{c1} = h^0_{c1}$, and the minimum of the spin wave mode 1 remains at $\mathbf{k}_1 = 0$. As a result, BEC condensation of mode 1 at $h = h_{c1}$ signals the transition to the commensurate Y state, which lives in the region OAF in Figure 5. In addition to breaking the continuous $U(1)$ symmetry, the Y state inherits broken Z_3 from the UUD phase (which corresponds to the selection of the sublattice for the *down* spins).

(**b**) For $1 < \delta \leq 4$, the low critical field increases and, simultaneously, the spin wave minima shift from zero to finite momenta $\pm\mathbf{k}_1 = (\pm k_1, 0)$. Thus, at $h = h_{c1}$, the spectrum softens at two different wave vectors at the same time. This

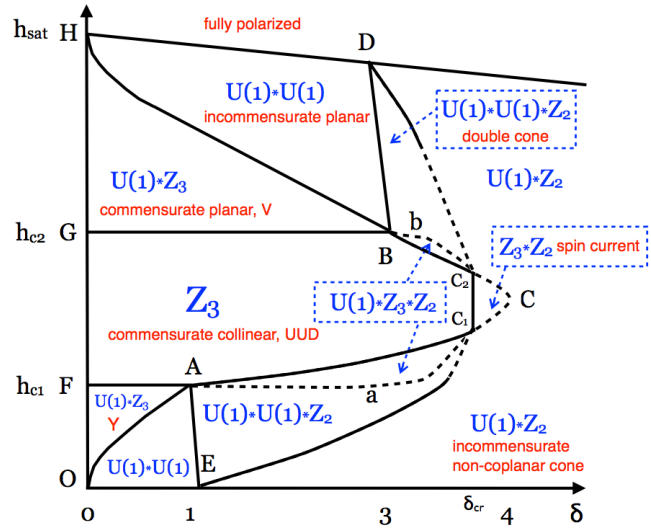

FIG. 5: Schematic large-S phase diagram of the deformed triangular lattice antiferromagnet in the $h - \delta$ plane, where h is magnetic field and deformation $\delta = (40/3)S(R/J)^2$. Phase boundary lines are schematic and serve to indicate possible shape only approximately. Dashed lines indicate location of conjectured phase transitions. Capital letters A through O indicated various (multi)critical points discussed in the text. Each phase is characterized by the order parameter manifold, indicated by (blue) symbols, which specifies symmetries broken by the phase relative to the fully symmetric (disordered) spin configuration. Most important for the review phases are additionally characterized by their key features/names, shown in red. Commensurate planar Y, V, collinear UUD, and non-coplanar cone state appearing in this diagram are sketched in Fig.2. Observe that the line H-B-b-C_2-C-C_1-a-A-O is the transition line separating phases with broken Z_3 from those with broken $U(1)$. This phase diagram summarizes results obtained in references 60–62.

opens an interesting possibility of the simultaneous coherent condensation of magnons with opposite momenta $+\mathbf{k}_1$ *and* $-\mathbf{k}_1$.[63] It turns out, however, that repulsive interaction between condensate densities at $\pm\mathbf{k}_1$ makes this energetically unfavorable[60]. Instead, at the transition the symmetry between the two possible condensates is broken spontaneously and magnons condense at a single momentum, $+\mathbf{k}_1$ *or* $-\mathbf{k}_1$. The resulting state, denoted as *distorted umbrella* in[60], is characterized by the broken $Z_3, U(1)$ *and* Z_2 symmetries – the latter corresponds to the choice $+\mathbf{k}_1$ or $-\mathbf{k}_1$. This non-coplanar state lives in the narrow region bounded by lines AC_1 (solid) and AaC_1 (dashed) in Figure 5.

(**c**) Similar developments occur near the upper critical field h_{c2}. In the interval $0 < \delta \leq 3$ the upper critical field is actually independent of δ, $h_{c2} = h^0_{c2}$, and the minimum of the spin wave mode 2 remains at $\mathbf{k}_2 = 0$. The transition on the line GB is towards the coplanar and commensurate V state, bounded by GBH in Figure 5. This state is characterized by broken $Z_3 \times U(1)$ symmetries.

(**d**) In the interval $3 < \delta \leq 4$, the upper field h_{c2} diminishes and simultaneously the minimum of the mode 2 shifts from zero momentum to the two degenerate locations at $\pm\mathbf{k}_2 = (\pm k_2, 0)$. Here, too, at the condensation transition

(along the line BC_2) it is energetically preferable to break Z_2 symmetry between the two condensates and to spontaneously select just one momentum, $+\mathbf{k}_2$ *or* $-\mathbf{k}_2$. This leads to *distorted umbrella* with broken $Z_3 \times U(1) \times Z_2$ symmetries. This state lives between (solid) BC_2 and (dashed) BbC_2 lines in Figure 5.

(**e**) *Spin nematic region.* The critical fields h_{c1} and h_{c2} merge at the plateau's end-point $\delta = 4$ (point C). The minima of the spin wave modes 1 and 2 coincide at this point $\mathbf{k}_1 = \mathbf{k}_2 = (k_0, 0)$ with $k_0 = \sqrt{3/(10S)}$. The end-point of the UUD phase thus emerges as a point of an extended symmetry hosting four linearly-dispersing gapless spin wave modes[60] (two branches, each gapless at $(\pm k_0, 0)$). Single-particle analysis of possible instabilities at the plateau's end-point (point C, $\delta = 4$) shows that in addition to the expected $U(1) \times U(1)$ symmetry (the two $U(1)$'s represent phases of the single particle condensates), the state at $\delta = 4$ point posses an unusual P_1 symmetry – the magnitude of the condensate at this point is not constrained[60]. The enhanced degeneracy of the plateau's end point C can also be understood from the observation that the two chiral distorted umbrella states merging at the point C are characterized, quite generally, by the *different* chiralities: Condensation of magnons at $h_{c1,c2}$, described in items (**b**) and (**c**) above, proceeds independently of each other - hence the chiralities of the two states are not related in any way. Thus the merging point of the two phases, point C, must possess enhanced symmetry.

Unconstrained magnitude mode hints at a possibility of a *two-particle condensation* - and indeed recent work[61] has found that a single-particle instability at $\delta = 4$ is pre-empted by the two-particle one at a slightly smaller value of $\delta = \delta_{\rm cr} = 4 - O(1/S^2)$. This is indicated by the (solid) line C_1-C_2 in Figure 5. This happens via the development of the 'superconducting'-like instability of the magnon pair fields $\Psi_{1,\mathbf{p}} = d_{1,+\mathbf{k}_0+\mathbf{p}} d_{2,-\mathbf{k}_0-\mathbf{p}}$ and $\Psi_{2,\mathbf{p}} = d_{1,-\mathbf{k}_0+\mathbf{p}} d_{2,+\mathbf{k}_0-\mathbf{p}}$ and consists in the appearance of two-particle condensates $\langle \Psi_{1,\mathbf{p}} \rangle = \langle \Psi_{2,\mathbf{p}} \rangle = i\Upsilon/|\mathbf{p}|$. The sign of the real-valued Ising order parameter Υ determines the sense of spin-current circulation on the links of the triangular lattice, as illustrated in Figure 6. The spin current is defined as the ground state expectation value of the vector product of neighboring spins. For example, spin current on the AC link of the elementary triangle is given by $\mathcal{J}^z_{AC} = \hat{z} \cdot \langle \mathbf{S}_A \times \mathbf{S}_C \rangle \sim \Upsilon$. In addition to spin currents, this novel state also supports finite scalar chirality, $\langle \mathbf{S}_A \cdot \mathbf{S}_B \times \mathbf{S}_C \rangle \sim \Upsilon$, even though $\langle S^{x,y}_{A/B/C} \rangle = 0$ for each of the spins individually. At the same time, in the absence of single-particle condensation, $\langle d_{1,2} \rangle = 0$, the usual two-point spin correlation function $\langle S^a(\mathbf{r}) S^b(0) \rangle$ is not affected by the two-particle $\langle \Psi_{1,2} \rangle \neq 0$ condensate: its transverse ($a = b = x$ or y) components continue to decay exponentially because of the finite energy gap in the single magnon spectra, while the longitudinal ($a = b = z$) components continue to show a perfect UUD crystal order.

Hence the resulting state, which lives inside triangle-shaped region C_1-C-C_2 in Figure 5, is a *spin-nematic* state. It can also be called a *spin-current* state[61]. It is uniquely characterized by the sign of Υ which determines the sense (clockwise or counterclockwise) of spin current circulation. The spin-current nematic is an Ising-like phase with massive excitations, which are domain walls separating domains of oppositely circling spin currents. It is characterized by broken $Z_3 \times Z_2$, where Z_2 is the sign of the two-magnon order parameter Υ in the ground state. It is useful to note that spontaneous selection of the circulation direction can also be viewed as a spontaneous breaking of the spatial inversion symmetry $\mathcal{I} : x \to -x$, which changes direction of spin currents on all bonds. (A different kind of nematic is discussed in Section IV B.)

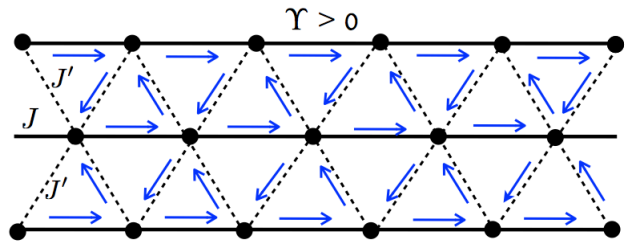

FIG. 6: Pattern of spin currents in the spin nematic phase, for $\Upsilon > 0$.

(**f**) *high-field region*, $h_{c2}(\delta) \ll h \leq h_{\rm sat}(\delta)$. The high field region, $h \approx h_{\rm sat}(\delta)$, can be conveniently analyzed within powerful Bose-Einstein condensation (BEC) framework. This follows from the simple observation that the ground state of the spin-S quantum model becomes a simple fully polarized state once the magnetic field is greater than the saturation field, $h > h_{\rm sat}(\delta)$ (which itself is a function of spin S and exchange deformation R). Excitations above this exact ground state are standard spin waves minimal energy for creating which is given by $h - h_{\rm sat}$. Similar to the situation near plateau's critical field $h_{c1/c2}$, these spin waves are characterized by non-trivial dispersion with two degenerate minima at momenta $\pm\mathbf{Q}'$ (see expression below (7)).

Spontaneous condensation in one of the minima, which constitutes breaking of Z_2 symmetry, results in the usual *cone*, or umbrella, state which is characterized by the spin pattern (7) which breaks spin-rotational $U(1)$. This state is realized to the right of the D-C_2 line in Figure 5.

Instead, simultaneous condensation of the high-field magnons in *both* minima results in the *coplanar* state[63] (also known as fan state) of the V type. For any $\delta \neq 0$ the wave vector $\mathbf{Q}'$ is not commensurate with the lattice, which makes the coplanar state to be *incommensurate* as well. The condensates $\Psi_{1,2} = \sqrt{\rho} e^{i\theta_{1,2}}$ at wave vectors $\pm\mathbf{Q}'$ have equal magnitude $\sqrt{\rho}$ and each breaks $U(1)$ symmetry. By choosing the phases $\theta_{1,2}$, the resulting state breaks *two* $U(1)$ symmetries. In total, the coplanar state is characterized by the broken $U(1) \times U(1)$. It is instructive to think of these symmetries in a slightly different way - as of those associated with the total $\theta_+ = \theta_1 + \theta_2$ and the relative $\theta_- = \theta_1 - \theta_2$ phases. The spin structure of this state is described by the incommensurate version of (5) which can be obtained by identifying $\varphi = \theta_+$ and replacing $\cos[\mathbf{Q} \cdot \mathbf{r}] \to \cos[\mathbf{Q}' \cdot \mathbf{r} + \theta_-]$. The latter replacement shows that the relative $U(1)$ symmetry, associated with the phase θ_-, can also be thought of as a *translational* symmetry associated with the shift of the spin configuration by the vector $\mathbf{r}_0$ such that $\mathbf{Q}' \cdot \mathbf{r}_0 = \theta_-$, modulo 2π.

Point H of the diagram, at $\delta = 0$ and $h = h_{\rm sat}$, is the special *commensurate* point. Here $\mathbf{Q}' \to \mathbf{Q} = (4\pi/3, 0)$, resulting in the commensurate coplanar V state. As noted right below Eq. (5), here $\mathbf{Q}\cdot\mathbf{r} = 2\pi\nu/3$, with $\nu = 0, 1, 2$, making the V state a three-sublattice one. Continuous $U(1)$ translational symmetry of the incommensurate V phase is replaced here by the discrete Z_3 symmetry. In other words, the relative phase θ_- is now restricted to the set of three equivalent (up to a global translation of the lattice) degenerate values. The $Z_3 \to U(1)$ transition across the line H-B between the two coplanar states is thus of a commensurate-incommensurate transition (CI) type. It is described by the classical two-dimensional sine-Gordon model with the nonlinear $\cos[3\theta_-]$ potential which locks the relative phase to the Z_3 set.[57] In the vicinity of point H in the diagram the line of the CI transition follows $h_{\rm sat} - h \sim \sqrt{\delta}$.[62] While the arguments presented here are valid in the immediate vicinity of $h_{\rm sat}$, the identification of the whole line H-B as the CIT line between the commensurate and incommensurate V states is possible due to the additional evidence reported in item (**c**) above – commensurate V state is reached from the UUD state in the whole interval $0 < \delta \le 3$.

The next task is to connect the incommensurate coplanar V state, which occupies region HDB in Figure 5, with the incommensurate cone state, to the right of D-C2 line. Since the V state has equal densities of bosons in the $\pm\mathbf{Q}'$ points, while the cone has finite density only in one of them, continuous transition between these two states at finite condensate density (that is, at any $h < h_{\rm sat}$) is *not possible*. At infinitesimally small condensate density, *i.e.* at $h = h_{\rm sat}$, direct transition is possible - it occurs at point D, which is a point of extended $U(1)\times U(1)\times U(1)$ symmetry: the two $U(1)$'s are phase symmetries while the third one is an emergent symmetry associated with the invariance of the potential energy at the constant total condensate density, $\rho = \rho_1 + \rho_2$, with respect to the distribution of condensate densities $\rho_{j=1,2} = \Psi_j^\dagger\Psi_j$ between the $\pm\mathbf{Q}'$ momenta. Large-S calculation of the ladder diagrams, which describe quantum corrections to the condensate energy, place point D at $\delta = 2.91$[62]. Assuming that the first order transition does not realize, we are forced to conclude that V and cone states must be separated by the intermediate phase, occupying DBbC$_2$ region. This phase breaks the $U(1)$ symmetry between ρ_1 and ρ_2 and interpolates smoothly between the symmetric situation $\rho_1 = \rho_2$, on the D-B line, and the asymmetric one $\rho = \rho_1$ and $\rho_2 = 0$ (or vice versa) on the line D-C$_2$. In doing so the momentum of the 'minority' condensate is found to evolve continuously from the initial $\mathbf{q}_2 = \mathbf{Q}'$ (which coincides with the momentum of the 'majority' condensate ρ_1) on the line D-C$_2$ to the final $\mathbf{q}_2 = -\mathbf{Q}'$ on the D-B line[62]. The resulting phase is a non-coplanar one, with strongly pronounced asymmetry in the $x-y$ plane: $\langle S^+_{\mathbf{r}}\rangle = \sqrt{\rho_1}e^{i\theta_1}e^{i\mathbf{Q}'\cdot\mathbf{r}} + \sqrt{\rho_2}e^{i\theta_2}e^{i\mathbf{q}_2\cdot\mathbf{r}}$. The state is characterized by the broken $U(1)\times U(1)\times Z_2$. For the lack of better term we call it *double cone*[62].

Going down along the field axis takes us toward the dashed B-b-C$_2$ line below which, according to the analysis summarized in item (c) above, represents a phase with broken $Z_3\times Z_2\times U(1)$. Hence along this line Z_3 is replaced by $U(1)$, which makes it a continuation of the C-IC transitions line H-B.

(**g**) *low-field region*, $0 \le h \ll h_{c1}(\delta)$. Semiclassical analysis at zero field $h = 0$ is well established and predicts incommensurate spiral state with zero total magnetization $M = 0$ of course. Quantum fluctuations renormalize strongly parameters of the spin spiral[64]. The most quantum case of the spin $S = 1/2$ remains not fully understood even for the relatively weak deformation of exchanges $R = J - J' \le J$ and is described in more details in Section III C.

At $\delta = 0$ one again has commensurate three-sublattice antiferromagnetic state, widely known as a 120° structure, which evolves into commensurate Y state in external magnetic field $h \ne 0$. Phenomenological analysis of Ref.57, Section III E, shows Y state becomes incommensurate when the deformation R exceeds $R_c \sim h^{3/2}$. Analysis near h_{c1}, reported in (**b**), tells that C-IC transition line must end up at point A. Comparing the energies of the incommensurate coplanar Y and the incommensurate umbrella state in the limit of vanishing magnetic field $h \to 0$, described in[60], identifies point E at $\delta = 1.1$ and $h = 0$ as the point of the transition between the $U(1)\times U(1)$ [the incommensurate coplanar V] and the $U(1)\times Z_2$ [the incommensurate cone] breaking states. Thus point E is analogous to point D.

By the arguments similar to those in part (**f**) above, there must be an intermediate phase with broken $U(1)\times U(1)\times Z_2$. It occupies region E-C$_1$-a-A in Figure 5. The state between A-C$_1$ and A-a-C$_1$ lines is characterized by different broken symmetries (see item (**b**) above) which makes the line A-a-C$_1$ to be the line of the $Z_3 \to U(1)$ transition. It thus has to be viewed as a continuation of the CIT line O-A.

To summarize, the large-S phase diagram in Figure 5 contains many different phases. It worth keeping in mind that it has been obtained under assumption of *continuous* phase transitions between states with different orders. Several of the phase boundaries shown in Fig. 5 are tentative – their existence is conjectured based on different symmetry properties of the states they are supposed to separate. To highlight their conjectured nature, such lines are drawn *dashed* in Figure 5. These include line B-b-C$_2$ which separates distorted umbrella (with broken $Z_3\times U(1)\times Z_2$) and double spiral (with broken $U(1)\times U(1)\times Z_2$), and similar to it line A-a-C$_1$ located right below the UUD phase. The end-points of these dashed lines are conjectured to be C$_2$ and C$_1$, correspondingly, which are the points of the two-magnon condensation [item (**e**)]. Line D-C$_2$, separating phases with broken $U(1)\times U(1)\times Z_2$ and $U(1)\times Z_2$, established via high-field analysis in item (**f**), is conjectured to end at the same C$_2$. Since different behavior cannot be ruled out at the present, its extension to the near-plateau region is indicated by the dashed line as well. Similar arguments apply to the line E-C$_1$. Finally, line C$_2$-C-C$_1$, covering the very tip of the UUD plateau phase, is made dashed because it is located past the two-magnon condensation transition (line C$_1$-C$_2$) into the spin-nematic state instabilities of which have not been explored in sufficient details yet.

One of the most unexpected and remarkable conclusions emerging from the analysis summarized here is the identification of the continuous line of C-IC transitions (line H-B-b-C$_2$-C-C$_1$-a-A-O), separating phases with discrete Z_3 from those

with continuous $U(1)$ symmetry. Its existence owes to the non-trivial interplay between geometric frustration and quantum spin fluctuations in the triangular antiferromagnet.

Spin excitation spectra: Many of the ordered *non-collinear* states described above harbor spin excitations with rather unusual characteristics. It has been pointed out some time ago[65,66] that local non-collinearity of the magnetic order results in the strong renormalization of the spin wave spectra at $1/S$ order. (This should be contrasted with the case of the collinear magnetic order, where quantum corrections to the excitation spectrum appear only at $1/S^2$ order.) This interesting effect, reviewed in[67], is responsible for dramatic flattening of spin wave dispersion, appearance of 'roton-like' minima, and significant shortening of spin wave lifetime. It also causes pronounced thermodynamic anomalies at temperatures as low as $J/10$.[68,69] Several recent experiments on 'large-spin' antiferromagnets α-$CaCr_2O_4$ ($S = 3/2$)[70,71] and $LuMnO_3$ ($S = 2$)[72] have observed these theoretically predicted features.

C. Spin 1/2 spatially anisotropic triangular antiferromagnet with $J' \neq J$

We now turn to the case of most quantum system: a spin $1/2$ antiferromagnet. Qualitatively, one expects quantum fluctuations to be most pronounced in this case, which suggests, in line with 'order-by-disorder' arguments of Section III A, a selection of the ordered Y, UUD and V states at and near the isotropic limit $J' \approx J$. Behavior away from this isotropic line represents a much more difficult problem, mainly due to the absence of physically motivated small parameter, which would allow for controlled analytical calculations. Aside from the two limits where small parameters do appear, namely the high field region near the saturation field and the limit of weakly coupled spin chains (see below), the only available approach is numerical.

Most of recent numerical studies of triangular lattice antiferromagnets focus on the zero field limit, $h = 0$, and on the phase diagram as a function of the ratio J'/J. These studies agree that a two-dimensional magnetic spiral order, well established at the isotropic $J' = J$ point, becomes incommensurate with the lattice when $J' \neq J$ and persists down to approximately $J' = 0.5J$. The ordering wave vector of the spiral $\mathbf{Q}'$ is strongly renormalized by quantum fluctuations[64,73,74] away from the semiclassical result. Below about $J' = 0.5J$, strong finite size effects and limited numerical accuracy of the exact diagonalization[75] and DMRG[73,76] methods do not allow one to obtain a definite answer about the ground state of the spin-1/2 $J - J'$ Heisenberg model, although most recent exact diagonalization studies employing twisted boundary conditions[74] and quantum fidelity analysis[77] have managed to access $J' \approx 0.2J$ region.

The apparent difficulty of reaching the true ground state of the model is a direct consequence of the strong frustration inherent in the triangular geometry. In the $J' \ll J$ limit the lattice decouples into a collection of linear spin chains weakly coupled by the frustrated interchain exchange $\mathcal{H}' = J' \sum_{x,y} \mathbf{S}_{x,y} \cdot (\mathbf{S}_{x-1/2,y+1} + \mathbf{S}_{x+1/2,y+1})$. Even classically, spin-spin correlations between spins from different chains are strongly suppressed as can be seen from the limit $J'/J \to 0$ when classical spiral wave vector $Q'_x = 2\cos^{-1}[-J'/2J] \to \pi + J'/J$. In this limit the relative angle between the spin at (integer-numbered) site x of the y-th chain and its neighbor at (half-integer-numbered) $x + 1/2$ site of the $y + 1$-th chain approaches $\pi/2 + J'/(2J)$. Thus the scalar product of two classical spins at neighboring chains vanishes as $J'/(2J) \to 0$.

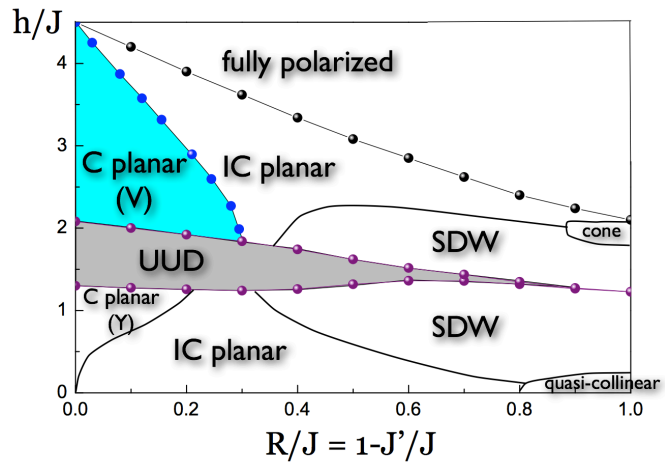

FIG. 7: Phase diagram of the spin $S = 1/2$ spatially anisotropic triangular antiferromagnet in magnetic field. Vertical axis - magnetic field h/J, horizontal - dimensionless degree of spatial anisotropy, $R/J = 1 - J'/J$. Notation C/IC stands for commensurate/incommensurate phases, accordingly. To compare this diagram with the large-S one in Fig.5, we note that collinear UUD and commensurate (C) planar Y and V phases, as well as their incommensurate (IC) planar versions (these are phases with the order parameter manifold $U(1) \times U(1)$ in Fig.5), are the same in the two diagrams. The narrow cone phase corresponds to incommensurate non-coplanar cone (with $U(1) \times Z_2$ manifold) in Fig.5. The SDW and quasi-collinear (which is a finite-field version of the described in the main text CoAF phase) phases have no analogues. Adapted from Ref.57.

Quantum spins add strong quantum fluctuations to this behavior, resulting in the numerically observed near-exponential decay of the inter-chain spin correlations, even for intermediate value $J'/J \lesssim 0.5$[73,74,76]. While some of the studies interpret such effective decoupling as the evidence of the spin-liquid ground state[75,76], the others conclude that the coplanar spiral ground state persists all the way to $J' = 0$[73,78].

Analytical renormalization group approach[79], which utilizes symmetries and algebraic correlations of low-energy degrees of freedom of individual spin-1/2 chains, finds that the system experiences quantum phase transition from the ordered spiral state to the unexpected collinear antiferromagnetic (CoAF) ground state. This novel magnetically ordered ground state is stabilized by strong quantum fluctuations of critical spin chains *and* weak fluctuation-generated interactions between next-nearest chains. This finding is supported by the coupled-cluster study[80,81], functional renormalization group[82] as well as by the combined DMRG and analytical

RG study[83]. Recent exact diagonalization study with twisted boundary conditions[74] have presented convincing evidence in favor of the CoAF state, while also noting that the competition between predominantly antiferromagnetic next-nearest chains correlations, which is a 'smoking gun' feature of the CoAF state[79,83], and predominantly ferromagnetic ones is "extremely close", resulting in the $\sim 10^{-7}$ difference of the appropriate dimensionless susceptibilities for the system of 32 spins[74]. In addition, variational wave function study of the spatially anisotropic $t - t' - U$ Hubbard model[84] also finds that quasi-one-dimensional collinear spin state is stable in a remarkably wide region of parameters, $t/t' \leq 0.8$ and $4 \leq U/t \leq 8$, see Sec. V C for more details.

Having described the limiting behavior along $h = 0$ and $J' = J$ ($R = 0$) axes, we now discuss the full $h - R$ phase diagram of the spin-1/2 Heisenberg model shown in Figure 7 ($R = J - J'$). The diagram is derived from extensive DMRG study of triangular cylinders (spin tubes) composed of 3, 6 and 9 chains and of lengths 120 - 180 sites (depending on R and the magnetization value) as well as detailed analytical RG arguments applicable in the limit $J' \ll J$[57]. It compares well, in the regions of small and intermediate R, with the only other comprehensive study of the full anisotropy-magnetic field phase diagram - the variational and exact diagonalization study by Tay and Motrunich[55]. The comparison is less conclusive in the regime of large anisotropy, $R \to 1$, which is most challenging for numerical techniques as already discussed above. (The biggest uncertainty of the diagram in Figure 7 consists in so far undetermined region of stability of the cone phase and, to a lesser degree, the phase boundaries between incommensurate (IC) planar and SDW phases.)

The main features of the phase diagram of the quantum spin-1/2 model are:

(1) High-field incommensurate coplanar (incommensurate V or fan) phase, which is characterized by the broken $U(1) \times U(1)$ symmetry, is stable for *all* values of the exchange anisotropy $1 \geq R \geq 0$. This novel analytical finding, confirmed in DMRG simulations, is described in Ref.57. This result is specific to the quantum $S = 1/2$ model – for any other value of the site spin $S \geq 1$ there is a quantum phase transition between the incommensurate planar and the incommensurate cone phases at some R_S. The critical value R_S is spin-dependent and decreases monotonically with S, from $R_S \approx 0.9$ for $S = 1$ to $R_S \approx 0.4$ for $S = 2$.

(2) 1/3 Magnetization plateau (UUD phase) is present for all values of R too: it extends from $R = 0$ all the way to $R = 1$. This striking conclusion is based on analytical calculations near the isotropic point[60], discussed in the previous Section, complementary field-theoretical calculations near the decoupled chains limit of $R \approx 1$[57,85] and on extensive DMRG studies of the UUD plateau in Ref.57.

(3) A large portion of the diagram in Figure 7, roughly to the right of $R = 0.5$, is occupied by the novel incommensurate collinear SDW phase. Physical properties of this magnetically ordered and yet intrinsically quantum state are summarized in Section IV A below.

Comparing the quantum phase diagram in Figure 7 with the previously described large-S phase diagram in Figure 5, one notices that both the high-field incommensurate coplanar and the UUD phases are present there too. The fact that both of these states become *more stable* in the spin-1/2 case and expand to the whole range of exchange anisotropy $0 < R < 1$, represents a striking *quantitative* difference between the large-S and $S = \frac{1}{2}$ cases. It should also be noticed that both phase diagrams demonstrate that the range of stability of the incommensurate cone (umbrella) phase is greatly diminished in comparison with that of the classical spin model.

In contrast, a collinear SDW phase, which occupies a good portion of the quantum phase diagram in Figure 7, is not present in Figure 5 at all - and this constitutes a major qualitative distinction between the large-S and the quantum $S = \frac{1}{2}$ cases.

IV. SDW AND NEMATIC PHASES OF SPIN-1/2 MODELS

A. SDW

The *collinear SDW phase* is characterized by the *modulated* expectation value of the local magnetization

$$\langle S^z_{\mathbf{r}} \rangle = M + \mathrm{Re}[\mathbf{\Phi} e^{i\mathbf{k}_{\mathrm{sdw}}\cdot\mathbf{r}}], \tag{8}$$

where $\mathbf{\Phi}$ is the SDW order parameter, and SDW wave vector $\mathbf{k}_{\mathrm{sdw}}$ is generally incommensurate with the lattice and, moreover, is the function of the uniform magnetization M and anisotropy R. Eq.(8) is very unusual for a classical (or semi-classical) spin system, where magnetic moments tend to behave as vectors of fixed length. It is, however, a relatively common phenomenon in itinerant electron systems with nested Fermi surfaces[86–88]. The appearance of such a state in a *frustrated* system of coupled spin-1/2 chains is rooted in a deep similarity between Heisenberg spin chain and one-dimensional spin-1/2 Dirac fermions[89–91]: thanks to the well-known phenomenon of one-dimensional spin-charge separation, the spin sectors of these two models are identical. Ultimately, it is this correspondence that is responsible for the 'softness' of the amplitude-like fluctuations underlying the collinear SDW state of Figure 7.

Figure 8 schematically shows a dispersion of $S = 1/2$ electron in a magnetic field. Fermi-momenta of up- and down-spin electrons $k_{\uparrow/\downarrow}$ are shifted from the Fermi momentum of non magnetized chain, $k_F = \pi/2$, by $\pm\Delta k_F = \pm\pi M$, where M is the magnetization. As a result, the momentum of the *spin-flip* scattering processes, which determine transverse spin correlation function $\langle S^+ S^- \rangle$, is given by $\pm(k_\uparrow - (-k_\downarrow)) = \pm 2k_F = \pm\pi$, and remains commensurate with the lattice. At the same time, longitudinal spin excitations, which preserve S^z, now involve momenta $\pm 2k_{\uparrow/\downarrow} = \pi(1 \pm 2M)$ and become incommensurate with the lattice. To put it differently, in the magnetized chain with $M \neq 0$ low-energy longitudinal spin fluctuations can be parameterized as $S^z(x) \sim \mathcal{S}^z e^{i(\pi - 2\pi M)x} + \mathcal{S}^{z*} e^{i(-\pi + 2\pi M)x}$, where (calligraphic) $\mathcal{S}^z(x)$ represents slow (low-energy) longitudinal mode. Similarly, transverse spin fluctuations are written as $S^+(x) \sim \mathcal{S}^+ e^{i\pi x}$, with $\mathcal{S}^+$ representing low-energy transverse mode.

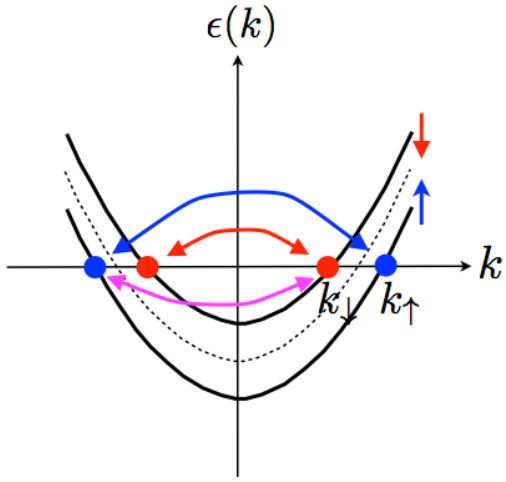

FIG. 8: Schematics of spinon dispersion for one-dimensional spin chain in magnetic field. Dashed line shows dispersion for zero field. $k_\uparrow(k_\downarrow)$ denote Fermi momentum of spin 'up' ('down') spinons, accordingly. Fermi momentum in the absence of the field is $k_F = \pi/2$.

This simple fact has dramatic consequences for frustrated interchain interaction $\mathcal{H}' = J' \sum_{x,y} \mathbf{S}_{x,y} \cdot (\mathbf{S}_{x-1/2,y+1} + \mathbf{S}_{x+1/2,y+1})$. Longitudinal ($z$) component of the *sum* of two neighboring spins on chain $(y+1)$ adds up to $S^z_{x-1/2,y+1} + S^z_{x+1/2,y+1} \to \sin[\pi M](\mathcal{S}^z_{x,y+1} e^{i(\pi - 2\pi M)x} + \text{h.c.})$, while the sum of their transverse components becomes a *derivative* of the smooth component of the transverse field $\mathcal{S}^+$, $S^+_{x-1/2,y+1} + S^+_{x+1/2,y+1} \to e^{i\pi x} \partial_x \mathcal{S}^+_{x,y+1}$. Hence the low-energy limit of the inter-chain interaction reduces to $\mathcal{H}' \to \sum_y \int dx \{ J' \sin[\pi M] \mathcal{S}^{z*}_{x,y} \mathcal{S}^z_{x,y+1} + J' \mathcal{S}^-_{x,y} \partial_x \mathcal{S}^+_{x,y+1} + \text{h.c.} \}$. The presence of the spatial derivative in the second term severely weakens it[85] and results in the domination of the density-density interaction (first term) over the transverse one (second term). The field-induced shift of the Fermi-momenta from its commensurate value, $k_F \to k_{\uparrow/\downarrow}$, together with frustrated geometry of inter-chain exchanges, are the key reasons for the field-induced stabilization of the two-dimensional longitudinal SDW state.

Symmetry-wise, the SDW state breaks no global symmetries (time reversal symmetry is broken by the magnetic field, which also selects the z axis), and, in particular, it preserves $U(1)$ symmetry of rotations about the field axis. This crucial feature implies the absence of the off-diagonal magnetic order $\langle S^{x,y}_{\mathbf{r}} \rangle = 0$ and would be gapless (Goldstone) spin waves. Instead, SDW breaks lattice translational symmetry. Its order parameter $\mathbf{\Phi} \sim \langle \mathcal{S}^z \rangle \neq 0$ is determined by inter-chain interactions[85,92]. Provided that $\mathbf{k}_{\text{sdw}} = \pi(1-2M)\hat{x}$ is *incommensurate* with the lattice, the only low energy mode is expected to be the pseudo-Goldstone acoustic mode of broken translations, known as a *phason*. (Inter-chain interactions do affect $\mathbf{k}_{\text{sdw}}$ but in the limit of small J'/J this can be neglected.) The phason is a purely longitudinal mode corresponding to the phase of the complex order parameter $\mathbf{\Phi}$ and hence represents a modulation of S^z only. This too is unusual in the context of insulating magnets, where, typically, the low energy collective modes are *transverse* spin waves, associated with small rotations of the spins away from their ordered axes. In the spin wave theory, longitudinal modes are typically expected to be highly damped[67,93,94], and hence hard to observe. (For a recent notable exception to this rule see a study of amplitude-modulated magnetic state of $PrNi_2Si_2$[95].) In the SDW state, the longitudinal phason mode in the only low energy excitation. Transverse spin excitations, which are also present in SDW, have finite energy gap. This, in fact, is one of key experimentally identifiable features of the SDW phase. More detailed description of spin excitations of this novel phase, as well as of the spin-nematic state reviewed below, can be found in the recent study[92].

At present, there are three known routes to the field-induced longitudinal SDW phase for a quasi-one-dimensional system of weakly coupled spin chains. The first, reviewed above, relies on the geometry-driven frustration of the transverse inter-chain exchange, which disrupts usual transverse spin ordering and promotes incommensurate order of longitudinal S^z components. The other route, described in the subsection IV A 1 below, relies on Ising anisotropy of individual chains. Lastly, it turns out that a two-dimensional SDW state may also emerge in a system of weakly coupled *nematic* spin chains - this unexpected possibility is reviewed in the subsection IV B.

1. *SDW in a system of Ising-like coupled chains*

There is yet another surprisingly simple route to the two-dimensional SDW phase. It consists in replacing Heisenberg chains with XXZ ones with pronounced Ising anisotropy. It turns out that a sufficiently strong magnetic field, applied along the z (easy) axis, drives individual chains into a critical Luttinger liquid state with dominant *longitudinal*, $S^z - S^z$, correlations[96]. This crucial property ensures that weak residual inter-chain interaction selects incommensurate longitudinal SDW state as the ground state of the anisotropic two-dimensional system.

We note, for completeness, that not every field-induced gapless spin state is characterized by the dominant longitudinal spin correlations. For example, another well-known gapped system, spin-1 Haldane chain, can also be driven into a critical Luttinger phase by sufficiently strong magnetic field[97–100]. However, that critical phase is instead dominated by strong transverse spin correlations[101,102]. As a result, a 2d ground state of weakly coupled spin-1 chains is a usual cone state[103].

2. *Magnetization plateau as a commensurate collinear SDW phase*

The *commensurate* case, when the wavelength $\lambda_{\text{sdw}} = 2\pi/k_{\text{sdw}}$ is a rational fraction of the lattice period, $\lambda_{\text{sdw}} = q/p$, requires special consideration. (Here the lattice period is set to be 1 and q and p are integer numbers.) Such commensurate state is possible at commensurate magnetization values

$M^{(p,q)} = \frac{1}{2}(1 - \frac{2q}{p})$. At these values, 'sliding' SDW state locks-in with the lattice, resulting in the loss of continuous translational symmetry. The SDW-plateau transition is then an incommensurate-commensurate transition of the sine-Gordon variety[57,85].

It turns out that in two-dimensional triangular lattice such locking is possible, provided that integers p and q have the same parity (both even or both odd)[85] (and, of course, provided that the spin system *is* in the two-dimensional collinear SDW phase). This condition selects $M = 1/3\ M_{\rm sat}$ plateau ($q = 1, p = 3$) as the most stable one, in a sense of the biggest energy gap with respect to creation of spin-flip excitation (which changes total magnetization of the system by ± 1). The next possible plateau is at $M = 3/5\ M_{\rm sat}$ ($q = 1, p = 5$)[85] – however this one apparently does not realize in the phase diagram in Figure 7, perhaps because it is too narrow and/or happen to lie inside the (yet not determined numerically) cone phase.

Applied to the one-dimensional spin chain, the above condition can be re-written as a particular $S = \frac{1}{2}$ version of the Oshikawa-Yamanaka-Affleck condition[104,105] for the period-p magnetization plateau in a spin-S chain, $pS(1 - M/M_{\rm sat}) =$ integer. Interestingly, this shows that $p = 3$ plateau at $M = 1/3\ M_{\rm sat}$ of the total magnetization $M_{\rm sat}$ is possible for all values of the spin S: the quantization condition becomes simply $2S =$ integer. This rather non-obvious feature has in fact been numerically confirmed in several extensive studies[106–108].

B. Spin nematic

Spin nematic represents another long-sought type of exotic ordering in the system of localized lattice spins. It breaks spin $SU(2)$ symmetry while preserving translational and time reversal symmetries. Out of many possible nematic states[109,110], our focus here is on bond-nematic order associated with the two-magnon pairing[111] and the appearance of the non-local order parameter $Q_{--} = S^-_{\mathbf{r}} S^-_{\mathbf{r}'}$ defined on the $\langle \mathbf{r}, \mathbf{r}' \rangle$ bond connecting sites $\mathbf{r}$ and $\mathbf{r}'$. Such order parameter can be build from quadrupolar operators $Q_{x^2-y^2} = S^x_{\mathbf{r}} S^x_{\mathbf{r}'} - S^y_{\mathbf{r}} S^y_{\mathbf{r}'}$ and $Q_{xy} = S^x_{\mathbf{r}} S^y_{\mathbf{r}'} + S^y_{\mathbf{r}} S^x_{\mathbf{r}'}$, as $Q_{--} = Q_{x^2-y^2} - iQ_{xy}$[112]. This bond-nematic order is possible in both $S \geq 1$, where quadrupolar order was originally suggested[113], and $S = 1/2$ systems of localized spins, coupled by exchange interaction.

The magnon pairing viewpoint, explored in great length in[37,112,114], is extremely useful for understanding basic properties of the spin-nematic state: the nematic can be thought of as a 'bosonic superconductor' formed as a result of two-magnon condensation $\langle S^-_{\mathbf{r}} S^-_{\mathbf{r}'} \rangle = \langle Q_{--} \rangle \neq 0$. As in a superconductor, a two-magnon condensate breaks $U(1)$ symmetry, which in this case is a breaking of the spin rotational symmetry with respect to magnetic field direction. It does not, however, break time-reversal symmetry (which requires a single-particle condensation). Just as in a superconductor, single-particle excitations of the nematic phase are gapped. This implies that transverse spin correlation function $\langle S^+_{\mathbf{r}} S^-_0 \rangle \sim e^{-r/\xi}$, which probes single magnon excitations, is short-ranged and decays exponentially. At the same time, fluctuations of magnon density, which are probed by longitudinal spin correlation function $\langle S^z_{\mathbf{r}} S^z_0 \rangle$, are sound-like acoustic (Bogoliubov) modes.

Before proceeding further, it is useful to distinguish spin nematic we are discussing here from other frequently discussed in the solid state literature nematic phases of itinerant electrons such as nematic Fermi-liquid[115] and (now experimentally confirmed) nematic phase of pnictide superconductors[116]. There, nematic correlations develop in the position space and correspond to a spontaneous selection of the preferred spatial direction (that is, spontaneous breaking of the lattice rotational symmetry) while still in the paramagnetic phase with no long-ranged magnetic order.

In contrast, the 'magnon-paired' state discussed here corresponds to a spontaneous selection of the direction in the internal spin space and is not sensitive to the often low, *e.g.* quasi-one-dimensional, lattice symmetries of the system.

1. *Weakly coupled nematic chains*

Basic ingredients of this 'bosonic superconductor' picture - gapped magnon excitations and attractive interaction between them - are nicely realized in the spin-1/2 quasi-one-dimensional material $LiCuVO_4$, reviewed in Section V B: the gap in the magnon spectrum is caused by the strong external magnetic field h which exceeds (single-particle) condensation field $h^{(1)}_{\rm sat}$, while the attraction between magnons is caused by the *ferromagnetic* (negative) sign of exchange interaction J_1 between the nearest spins of the chain (see Figure 9). Under these conditions, the two-magnon bound state, which lies below the gapped single particle states, condenses at $h^{(2)}_{\rm sat}$, which is higher than $h^{(1)}_{\rm sat}$. As a result, a spin nematic state is naturally realized in each individual chain in the intermediate field interval $h^{(1)}_{\rm sat} < h < h^{(2)}_{\rm sat}$.[114]

Note, however, that a true $U(1)$ symmetry breaking is not possible in a single chain, where instead a critical Luttinger state with algebraically decaying nematic correlations is established[112]. To obtain a true *two-dimensional nematic* phase, one needs to establish a phase coherence between the phases of order parameters $Q_{--}(y)$ of different chains. By our superconducting analogy, this requires an inter-chain Josephson coupling. It transfers bound two-magnon pairs between nematic chains. The corresponding 'pair-hopping' reads $K \sum_{x,y} (Q_{++}(x,y) Q_{--}(x,y+1) + \text{h.c.})$. Microscopically, such an interaction represents a four-spin coupling, which is not expected to be particularly large in a good Mott insulator with a large charge gap, such as $LiCuVO_4$. However, even if K is absent microscopically, it will be generated perturbatively from the usual inter-chain spin exchange $J' \sum_{x,y} (S^+_{x,y} S^-_{x,y+1} + \text{h.c.})$, which plays the role of a single-particle tunneling process in the superconducting analogy. (Observe that expectation value of this interaction in the chain nematic ground state is zero – adding or removing of a single magnon to the 'superconducting magnon' ground state is forbidden at energies below the single magnon gap.) The pair-tunneling generated by fluctuations is estimated to be of the

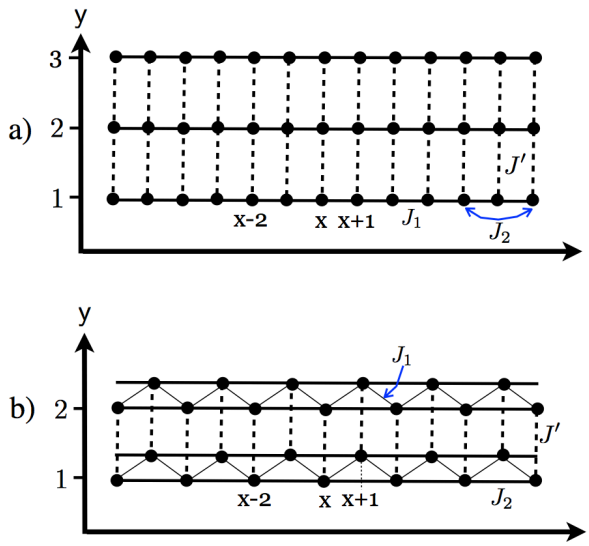

FIG. 9: (a) Weakly coupled nematic chains with ferromagnetic J_1 (solid lines) and antiferromagnetic next-nearest J_2, coupled by an inter-chain exchange J'. (b) Same system viewed as a set of weakly coupled zig-zag chains.

order $K \sim (J')^2/J_1 \ll J'$.

At the same time, $S^z - S^z$ interaction between chains does not suffer from a similar 'low-energy suppression'. This is because S^z is simply proportional to a number of magnon pairs, $n_{\rm pair}(x,y) = 2n(x,y)$, which is just twice the magnon number $n(x,y)$. Hence $S^z(x,y) = 1/2 - 2n(x,y)$ differs only a by coefficient 2 from its usual expression in terms of magnon density.

We thus have a situation where the strength of interchain density-density ($S^z - S^z$) interaction, which is determined by the original interchain J', is much stronger than that for the fluctuation-generated Josephson interaction $K \sim (J')^2/J_1$. In addition, more technical analysis of the scaling dimensions of the corresponding operators shows[92] that the two competing interactions are characterized by (almost) the same scaling dimension (approximately equal to 1 for $h \approx h_{\rm sat}$) which makes them both strongly relevant in the renormalization group sense. Given an inequality $J' \gg (J')^2/J_1$, which selects interchain $S^z - S^z$ interaction as the strongest one, we end up with a *two-dimensional collinear SDW phase* built out of nematic spin chains[92,117]. This conclusion holds for all h except for the immediate vicinity of the saturation field $h^{(2)}_{\rm sat}$. There a separate fully two-dimensional BEC analysis is required, due to the vanishing of spin velocity at the saturation field, and the result is a true *two-dimensional nematic phase* in the narrow field range $h^{(1)}_{\rm sat} \lesssim h \leq h^{(2)}_{\rm sat}$[92,114,117]. A useful analogy to this competition is provided by models of striped superconductors, where the competition is between the superconducting order (a magnetic analogue of which is the spin nematic) and the charge-density wave order (a magnetic analogue of which is the collinear SDW), see Ref.118 and references therein.

To summarize, weak inter-chain interaction J' between $J_1 - J_2$ spin chains with strong nematic spin correlations actually stabilizes a two-dimensional SDW phase as the ground state in a wide range of magnetization. This state preserves $U(1)$ symmetry of spin rotations and is characterized by short-ranged transverse spin correlations, similar to a nematic state.

2. *Spin-current nematic state at the 1/3-magnetization plateau*

The discussion in the previous Subsection was focused on the systems with ferromagnetic exchange ($J_1 < 0$) on some of the bonds - as described there, negative exchange needed in order to obtain an *attractive* interaction between magnons.

Extensively used above superconducting 'intuition' forces one to ask, by analogy with superconducting states of repulsive fermion systems (such as, for example, pnictide superconductors or high-temperature cuprate ones), if it is possible to realize a spin-nematic in a spin system with only antiferromagnetic (that is, repulsive) exchange interactions between magnons. To the best of our knowledge, the first example of such a state is provided by the spin-current state described in part (**e**) of the Section III B. Being nematic, this state is characterized by the chiral (spin current) long-range order and the absence of the magnetic long-range order in the transverse to the magnetic field direction[61], as sketched in Fig.6.

A very similar state, named *chiral Mott insulator*, was recently discovered in the variational wave function study of a two-dimensional system of interacting bosons on frustrated triangular lattice[119] as well as in a one-dimensional system of bosons on frustrated ladder[120,121]. In both cases, a chiral Mott insulator is an intermediate phase, which separates the usual Mott insulator state (which is a boson's analogue of the UUD state) from the superfluid one (which is an analogue of the cone state). As in Figure 5, it intervenes between the states with distinct broken symmetries (Z_3 and $U(1) \times Z_2$ in our case), and gives rise to two continuous transitions instead of a single discontinuous one.

C. Magnetization plateaus in itinerant electron systems

Up-up-down magnetization plateaus, found in the triangular geometry, are of classical nature. Over the years, several interesting suggestions of non-classical (liquid-like) magnetization plateaux have been put forward[122–127], but so far not observed in experiments or numerical simulations. Very recently, however, two numerical studies[128,129] of the spin-1/2 kagome antiferromagnet have observed non-classical magnetization plateaus at $M/M_{\rm sat} = 1/3, 5/9, 7/9$. These intriguing findings, taken together with earlier prediction of a *collinear spin liquid* at $h = h_{\rm sat}/3$ in the classical kagome antiferromagnet[130], hint at a very rich magnetization process of the quantum model, the ground state of which at $h = 0$ is a Z_2 spin liquid![4].

A different point of view on the magnetization plateau was presented recently in Ref.131. The authors asked if the plateau is possible in an itinerant system of weakly-interacting electrons. The answer to this question is affirmative, as can be

understood from the following consideration.

Let us start with a system of non-interacting electrons on a triangular lattice. A magnetic field, applied in-plane in order to avoid complications due to orbital effects, produces magnetization $M = (n_\uparrow - n_\downarrow)/2$, where densities n_σ of electrons with spin $\sigma =\uparrow,\downarrow$ are constrained by the total density $n = n_\uparrow + n_\downarrow$. Consider now special situation with $n_\uparrow = 3/4$, at which the Fermi-surface of $\sigma =\uparrow$ electrons, by virtue of lattice geometry, acquires particularly symmetric shape: a hexagon inscribed inside the Brillouin zone hexagon, see Figure 10.

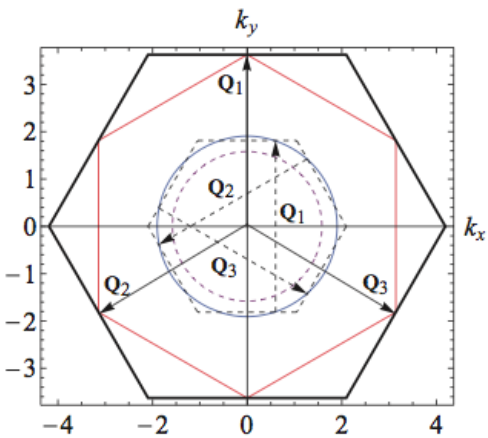

FIG. 10: The Fermi surface (red hexagon) of the non-interacting $\sigma =\uparrow$ electrons on a triangular lattice at $n_\uparrow = 3/4$. Nearly circular Fermi surface of the minority $\sigma =\downarrow$ electrons is shown by thin (blue) line. Bold black hexagon represents the Brillouin zone. Adapted from Zhihao Hao and Oleg A. Starykh, Phys. Rev. B 87, 161109 (2013).

Points where $\sigma =\uparrow$ Fermi-surface touches the Brillouin zone (denoted by vectors $\pm\mathbf{Q}_j$ with $j = 1, 2, 3$ in Fig.10) are the van Hove points, at which Fermi-velocity vanishes and electron dispersion becomes quadratic. They are characterized by the logarithmically divergent density of states. In addition, being a hexagon, $\sigma =\uparrow$ Fermi-surface is perfectly nested. As a result, static susceptibility $\chi_\uparrow(\mathbf{k})$ of spin-up electrons is strongly divergent, as $\log^2(|\mathbf{k}-\mathbf{Q}_j|)$, for wave vectors $\mathbf{k} \approx \mathbf{Q}_j$.

Given this highly susceptible spin-$\uparrow$ Fermi surface, it is not surprising that a weak interaction between electrons, either in the form of a direct density-density interaction $V n_{\mathbf{r}} n_{\mathbf{r}+\delta_j}$ between electrons on, e.g., nearest sites, or in the form of a local Hubbard interaction $U n_{\mathbf{r},\uparrow} n_{\mathbf{r},\downarrow}$ between the majority and minority particles, drives spin-$\uparrow$ electrons into a gapped correlated state - the charge density wave (CDW) state[131] with fully gapped Fermi surface. Moreover, CDW ordering wave vectors $\mathbf{Q}_j$ are commensurate with the lattice, leading to a commensurate CDW for the spin-$\uparrow$ electrons[132].

Minority spin-$\downarrow$ electrons experience position-dependent effective field $U\langle n_{\mathbf{r},\uparrow}\rangle$ and form CDW as well. Depending on whether or not vectors $\mathbf{Q}_j$ span the spin-down Fermi-surface (this depends on $n_\downarrow$ density), the Fermi-surface of $\sigma =\downarrow$ electrons may or may not experience reconstruction. However, being not nested, it is guaranteed to retain at least some parts of the critical Fermi-surface.

The resulting state is a co-existence of a charge- and collinear spin-density waves, together with critical $\sigma =\downarrow$ Fermi surface. Since the energy cost of promoting a spin-$\downarrow$ electron to a spin-$\uparrow$ state is finite (and given by the gap on the spin-$\uparrow$ Fermi surface), the resulting state realizes *fractional* magnetization plateau, the magnetization of which is determined by the total density n via $M = (3/2 - n)/2$. At half-filling, $n = 1$, the plateau is at $1/2$ of the total magnetization, but for $n \neq 1$ it takes a fractional value. Amazingly, the obtained state is also a *half-metal*[133] - the only conducting band is that of (not gapped) minority spin-$\downarrow$ electrons.

Theoretical analysis sketched here bears strong similarities with recent proposals[132,134–137] of collinear and chiral spin-density wave (SDW) and superconducting states of itinerant electrons on a honeycomb lattice in the vicinity of electron filling factors 3/8 and 5/8 in zero magnetic field. Our analysis shows that even simple square lattice may host similar half-metallic magnetization plateau state[131]. Similar to the case of a magnetic insulator, described in the previous Sections, external magnetic field sets the direction of the collinear CDW/SDW state. The resulting half-metallic state only breaks the discrete translational symmetry of the lattice, resulting in fully gapped excitations, and remains stable to fluctuations of the order parameter about its mean-field value. In addition to standard solid state settings, the described phenomenon may also be observed in experiments on cold atoms, where desired high degree of polarization can be easily achieved[138]. It appears that, in addition to the half-metallic state, the system may also support p-wave superconductivity - a competition between these phases may be efficiently studied with the help of functional renormalization group[139].

V. EXPERIMENTS

Much of the current theoretical interest in quantum antiferromagnetism comes from the amazing experimental progress in this area during the last decade. The number of interesting materials is too large to review here, and for this reason we focus on a smaller sub-set of recently synthesized quantum spin-1/2 antiferromagnets, which realize some of quantum states discussed above.

One of the best known among this new generation of materials is Cs_2CuCl_4, extensively studied by Coldea and collaborators in a series of neutron scattering experiments[20,140,141] and by others via NMR[142] and, more recently, ESR[143–147] experiments. This spin-1/2 material represents a realization of a deformed triangular lattice with $J'/J = 0.34$[20] and significant DM interactions on chain and inter-chain (zig-zag) bonds, connecting neighboring spins[20,85]. Inelastic neutron scattering experiments have revealed unusually strong multi-particle continuum, the origin of which has sparked intense theoretical debate[148–155]. The current consensus is that Cs_2CuCl_4 is best understood as a weakly-ordered quasi-one-dimensional

antiferromagnet, whose spin excitations smoothly interpolate from fractionalized spin-1/2 spinons of one-dimensional chain at high- and intermediate energies to spin waves at lowest energy ($\ll J'$)[155]. Although weak, residual inter-plane and DM interactions play the dominant role in the magnetization process of this material. The resulting $B-T$ phase diagram is rather complex and highly anisotropic[156], and does not contain a magnetization plateau. However it is worth mentioning that this was perhaps the first spin-1/2 material, a magnetic response of which featured a SDW-like phase ordering wave vector, which scales *linearly* with magnetic field in an about 1 tesla wide interval (denoted as phase "S" in[140] and as phase "E" in[156]). While still not well understood, this experimental observation have provided valuable hint to quasi-1d approach based on viewing Cs_2CuCl_4 as a collection of weakly coupled spin chains[155].

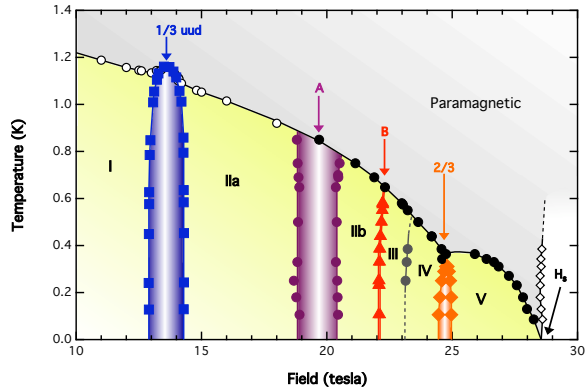

FIG. 11: Magnetic phase diagram of Cs_2CuBr_4, as deduced from the magnetocaloric-effect data taken at various temperatures. Circles indicate second-order phase boundaries, whereas other symbols except the open diamonds indicate first-order boundaries. Adapted from Fortune *et al.*, Phys. Rev. Lett. 102, 257201 (2009).

A. Magnetization plateau

Robust $1/3$ magnetization plateau – the first of its kind among triangular spin-1/2 antiferromagnets – is present in Cs_2CuBr_4, which has the same crystal structure as Cs_2CuCl_4, but is believed to be more two-dimensional-like in a sense of having higher J'/J ratio. What that ratio is however a subject of current debates. By fitting the experimentally determined ordering wave vector $\mathbf{q}_0$ of the incommensurate helical state in zero magnetic field to the predictions of the classical spin model one obtains $J'/J = 0.47$[157]. When fit to the predictions of the $S = 1/2$ series expansion study[64], the same $\mathbf{q}_0$ results in a much bigger estimate $J'/J \approx 0.74$[157]. High-temperature series fit of the uniform susceptibility results in $J'/J \approx 0.5$.[158] Recent high-field ESR experiment, which probes magnetic excitation spectra of the magnons of the fully polarized state, gives $J'/J \approx 0.41$[147]. Importantly, the same ESR experiment, when done on Cs_2CuCl_4, predicts $J'/J \approx 0.3$[147], in a close agreement with the estimate $J'/J = 0.34$, already quoted above and derived from earlier neutron scattering experiment[20]. It thus appears that $J'/J \approx 0.5$ should be accepted as an estimate of the exchange ratio for Cs_2CuBr_4.

The observed plateau, which is about 1 tesla wide ($h_{c1} = 13.1$T and $h_{c2} = 14.4$T)[157,159,160], is clearly visible in both magnetization and elastic neutron scattering measurements[157,160], which determined the UUD spin structure on the plateau. The observation of the magnetization plateau has generated a lot of experimental activity. The quantum origin of the plateau visibly manifests itself via essentially temperature-independent plateau's critical fields $h_{c1,2}(T) \approx h_{c1,2}(T=0)$, as found in the thermodynamic study[161]. This behavior should be contrasted with the phase diagram of the spin-5/2 antiferromagnet $RbFe(MoO_4)_2$[30,162,163], where the critical field $h_{c1}(T)$ does show strong downward shift with T. (Recall that in the classical model, Figure 3, the UUD phase collapses to a single point at $T = 0$.)

Commensurate up-up-down spin structure of Cs_2CuBr_4 is also supported by NMR measurements[164,165] which in addition find that transitions between the commensurate plateau and adjacent to it incommensurate phases are discontinuous (first-order). Extensive magnetocaloric effect and magnetic-torque experiments[166] have uncovered surprising cascade of field-induced phase transitions in the interval $10-30$ T. The most striking feature of the emerging complex phase diagram, shown in Figure 11, is that it appears to contain up to 9 different magnetic phases – in stark contrast with the 'minimal' theoretical model diagram in Figure 3 which contains just 3 phases! This, as well as strong sensitivity of the magnetization curve to the direction of the external magnetic field with respect to crystal axis, strongly suggest that the difference in the phase diagrams has to do with spatial ($J' \neq J$) and spin-space (asymmetric DM interaction) anisotropies present in Cs_2CuBr_4. Large-S and classical Monte Carlo studies[31] do find the appearance of new incommensurate phases in the phase diagram, in qualitative agreement with the large-S diagram of Figure 5 (note that the latter does not account for the DM interaction which significantly complicates the overall picture[31]).

Perhaps the most puzzling of the "six additional" phases is a narrow region at about $B = 23$T, where dM/dB exhibits sharp double peak structure, interpreted in[157,167] as a novel magnetization plateau at $M/M_{\rm sat} = 2/3$. Such a new $2/3$-magnetization plateau was observed in an exact diagonalization study of spatially anisotropic spin-1/2 model[168] but was not seen in more recent variational wave function[55] and DMRG[57], as well as in analytical large-S[31,60] studies.

Nearly isotropic, $J'/J \approx 1$, antiferromagnet $Ba_3CoSb_2O_9$ is believed to provide an 'ideal' realization of the spin-1/2 antiferromagnet on a uniform triangular lattice[21,169]. And, indeed, its experimental phase diagram is in close correspondence with $J' = J$ cut in Figure 5 (along $\delta = 0$ line) and Figure 7 (along $R = 0$ line): it has 120° spin structure at zero field, coplanar Y state at low fields, the 1/3 magnetization plateau in the $h_{c1}/h_{\rm sat} = 0.3 \leq h/h_{\rm sat} \leq h_{c2}/h_{\rm sat} = 0.47$ interval, and coplanar V state (denoted as $2:1$ state in[169]) at higher fields.

A new element of the study[21,169] is the appearance of weak

anomaly in dM/dB at about $M/M_{\rm sat} = 3/5$, which was interpreted as a quantum phase transition from the V phase to another coplanar phase - inverted Y (state 'e' in Figure 2). Near the saturation field these two phases are very close in energy, the difference appears only in the 6th order in condensate amplitude[63]. Such a transition can be driven by sufficiently strong easy-plane anisotropy[62,170] as well as anisotropic DM interaction[31]. (Note, however, that recent DMRG study[171] claims that the inverted Y state is a finite-size effect which is not present in the thermodynamic limit of the spin-$1/2$ problem.)

However, perhaps a more likely explanation is that the transition is due to the weak interlayer exchange interaction. Ref. 172 has shown that weak inter-plane antiferromagnetic exchange interaction causes transition from the uniform V phase to the *staggered V* phase. The latter is described by the same Eq.(5) but with a z-dependent phase, $\varphi_z = \tilde{\varphi} + \pi z$ (here, z is the integer coordinate of the triangular layer and $\tilde{\varphi}$ is an overall constant phase), leading to the doubling of the period of the magnetic structure along the direction normal to the layer. It is easy to see that such a state actually gains energy from the antiferromagnetic interlayer exchange J'', while preserving the optimal in-plane configuration in every layer. Such a transition, denoted as HFC1-HFC2 transition, was also observed in recent semi-classical Monte Carlo simulations[173]. This development suggest that a mysterious '2/3-plateau' of Cs_2CuBr_4, mentioned above, may also be related to a transition between the lower-field uniform and higher-field staggered versions of the commensurate V phase.

B. SDW and spin nematic phases

An "Ising" route to the collinear SDW order, described in Section IV A 1, has been realized in spin-1/2 Ising-like antiferromagnet $BaCo_2V_2O_8$. Experimental confirmations of this comes from specific heat[174] and neutron diffraction[175] measurements. The latter one is particularly important as it proves the linear scaling of the SDW ordering wave vector with the magnetization, $k_{\rm sdw} = \pi(1-2M)$, predicted in[176]. Subsequent NMR[177], ultrasound[178], and neutron scattering[179] experiments have refined the phase diagram and even proposed the existence of two different SDW phases[177] stabilized by competing interchain interactions.

Another Co-based quasi-one-dimensional material, $Ca_3Co_2O_6$, represents an outstanding puzzle. This strong $S = 2$ Ising ferromagnet shows both a $1/3$ magnetization plateau and an incommensurate longitudinal SDW ordering below it[180–183], which, however, do not seem to represent its true equilibrium low-temperature phases: the material exhibits strong history dependence[183] and associated with it extremely slow, on a scale of several hours, temporal evolution[184] towards an ultimate equilibrium ground state. This, in the case of no external magnetic field, appears to be a commensurate antiferromagnetic phase. Refs.[183,184] have argued that strong Ising character of spin-2 chains, as estimated by the large ratio of the easy-axis anisotropy $D \approx 110$K to the in-chain ferromagnetic exchange constant $J_1 \approx 6$K, is the reason for this extremely slow dynamics – very weak quantum fluctuations lead to a very small relaxation rate.

Most recently, spin-1/2 magnetic insulator $LiCuVO_4$ has emerged[185,186] as a promising candidate to realize both a high-field spin nematic phase, right below the two-magnon saturation field, which is at about 45 tesla high, and an incommensurate collinear SDW phase at lower fields, extending from about 40T down to about 10 T. At yet lower magnetic field, the material realizes more conventional vector chiral (umbrella) state which can be stabilized by a moderate easy-plane anisotropy of exchange interactions[187] (which does not affect the high field physics discussed here).

This material seems to nicely realize theoretical scenario outlined in Section IV B 1: spin-nematic chains[188,189] form a two-dimensional nematic phase only in the immediate vicinity of the saturation field[190,191]. (Apparently, more detailed analysis of this near-saturation region also requires disentangling contribution of the nonmagnetic vacancies from that of the 'ideal' spin system, which is not an easy task at all[191,192].) At fields below that rather narrow interval, the ground state is an incommensurate longitudinal SDW state. Evidence for the latter includes detailed studies of NMR line shape[193–196] and neutron scattering[197,198]. It is worth adding here that quasi-one-dimensional nature of this material is evident from the very pronounced multi-spinon continuum, observed at $h = 0$ in inelastic neutron scattering studies[199].

C. Weak Mott insulators: Hubbard model on anisotropic triangular lattice

Given that, quite generally, Heisenberg Hamiltonian can be viewed as a strong-coupling (large U/t) limit of the Hubbard model, it is natural to consider the fate of the Hubbard $t - t' - U$ model on (spatially anisotropic, in general) triangular lattice.

As a matter of fact, this very problem is of immediate relevance to intriguing experiments on organic Mott insulators of $X[Pd(dmit)_2]_2$ and κ-$(ET)_2Z$ families. Recent experimental[8,22] and theoretical[12,200–202] reviews describe key relevant to these materials issues, and we direct interested readers to these publications.

One of the main unresolved issues in this field is that of a proper minimal model that captures all relevant degrees of freedom. Highly successful initial proposal[205] models the system as a simple half-filled Hubbard model on a spatially anisotropic triangular lattice, as sketched in Figure 12. Every 'site' of this lattice in fact represents closely bound dimer, made of two ET molecules[205], and occupied by single electron (or hole). Spatial anisotropy shows up via different hopping integrals t, t'. Within the standard large-U description, where $J = 2t^2/U$ for the given bond, anisotropy of hopping t'/t directly translates into that of exchange interactions on different bonds, $J'/J \sim (t/t')^2$. Ironically, most of the studied materials fall onto $t' < t$ side[8,203] of the diagram, which happens to be opposite to $J > J'$ limit of spatially anisotropic Heisenberg model, to which this review is devoted.

This description has generated a large number of interest-

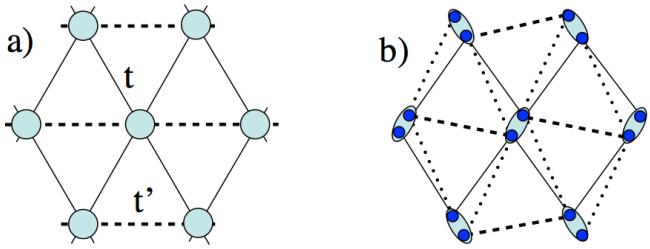

FIG. 12: (a) Spatially anisotropic single-band Hubbard model for organic Mott insulators, after Refs.8,203. Note that assignment of the primed and non-primed bonds of the lattice, within the Hubbard model nomenclature which is used in this Figure and the whole of Sec.V C, is opposite to that in the Heisenberg model used in Fig. 1 and all other sections of the review. Thus, $t'/t < 1$ here implies $J/J' < 1$ in Figure 1: J' of Fig. 1 actually lives on t bonds of the Hubbard model here. (b) Geometry of an extended two-band model, after Ref.204. The 'sites', shown by filled dots inside ovals representing dimers, are now two-molecule dimers. Two sites from the same dimer are connected by the intra-dimer tunneling amplitude t_d, and in the limit $t_d \gg$ 'inter-dimer tunnelings' the model reduces to that in (a). Note also the appearance of additional hopping amplitudes (dotted line) connecting 'more distant' sites of neighboring dimers.

ing studies, the full list of which is beyond the scope of this review and most of which are again on the 'wrong' side of the t/t' ratio. One notable exception is provided by Ref.84, which represents an extensive study of the $t/t' - U$ phase diagram of the Hubbard model on spatially anisotropic triangular lattice, see Figure 2 of that paper. Interestingly, this phase diagram contains an extended region of the large-U spin-liquid phase (which occupies squarish region of $0 < t/t' < 0.8$ and $U/t' \geq 10$), which is separated from the metallic low-U phase (which occupies a wedge of $0 < t/t' < 0.8$ and $0 < U/t' \leq 5$) by the extended region of magnetically ordered collinear phase. This unexpected phase appears to be similar to the CoAF state of the $J-J'$ model described in Section III C. Its static spin structure factor peaks at $k_x = \pi$ and is very much insensitive of the value of k_y, suggesting strong one-dimensional character of the state. This is further confirmed by the analysis of renormalized 'Fermi surfaces' (see Ref.84 for explanations) which also show very pronounced one-dimensional feature of being very weakly modulated as a function of transverse component of the momentum k_y. Taken together with the fact that the spin liquid state is strongly one-dimensional as well[84], these findings suggest that the whole region of $0 < t/t' \leq 0.8$ and $U/t' \geq 5$ is adequately described by the model of weakly coupled Hubbard chains - and that, similar to what happens in the spatially anisotropic triangular antiferromagnet with $J' < J$, Sec. III C, it is the frustrated (zig-zag) geometry of the interchain hopping t which is responsible for the strong 'dimensional reduction'[12,79], observed numerically in Ref.84.

Let us now turn to a more experimentally relevant regime of $t/t' > 1$. We start by focusing on two materials from κ-(ET)$_2$Z family: κ-(ET)$_2$Cu[N(CN)$_2$]Cl (abbreviated as κ-Cl in the following) and κ-(ET)$_2$Cu$_2$(CN)$_3$ (abbreviated as κ-CN). At low temperature and ambient pressure, the first of these, κ-Cl, is an ordered antiferromagnetic insulator, while κ-CN is a Mott insulator showing no signs of magnetic order down to lowest temperature[7,206]. Band structure calculations, based on extended Hückel method, have predicted[207] $t'/t = 0.75$ for κ-Cl and $t'/t = 1.06$ for κ-CN. Modern density functional calculations, however, predict[208,209] quite different ratios: $t'/t = 0.4$ for κ-Cl and $t'/t = 0.83$ for κ-CN. Strong downward revision of the t'/t ratio has equally strong affect on the ratio of the exchanges: $J'/J \approx 6$ for κ-Cl and $J'/J \approx 1.5$ for κ-CN, suggesting that both materials can be viewed as weakly distorted square lattice antiferromagnets (the square lattice is formed by J' bonds). But this is cannot be the complete picture as we do know that κ-CN does not magnetically order. At the same time, high-temperature series expansion for the uniform susceptibility[158] is very much consistent with that of the *uniform* triangular lattice antiferromagnet with exchange constant of about 250K - this however cannot be a full story either as the triangular lattice antiferromagnet has an ordered ground state as well!

Clearly, Heisenberg-type spin-only description of these interesting materials is not sufficient for explaining their puzzling magnetically-disordered Mott insulator behavior. It is important to realize that for not too large U/t, the standard Heisenberg model must be amended with *ring-exchange* terms involving four (or longer) loops of spin operators[210,211]. This addition dramatically affects the regime of intermediate U/t by stabilizing an insulating spin-liquid ground state[211,212]. The nature of the emerging spin-liquid is subject of intense on-going investigations, with proposals ranging from Z_2 liquid[213,214] to spin Bose-metal[215], to spin-liquid with quadratic band touching[216].

Recently, however, this appealing spin-only picture of the organic Mott insulators has been challenged by the experimental discovery of anomalous response of dielectric constant[217,218] and lattice expansion coefficient[219] at low temperature. This finding imply that charge degrees of freedom, assumed frozen in the spin-only description, are actually present in the material and have to be accounted for in the theoretical modeling. Several subsequent papers[204,220–222] have identified dimer units of the triangular lattice, viewed as sites in Figure 12, as the most likely place where charge dynamics persists down to lowest temperatures. To describe these internal states of the two-molecule dimers, one need to go back to a two-band extended Hubbard model description[223]. (Note that magneto-electric coupling is in fact also present in a sinlge-band band Hubbard model, where it appears in certain frustrated geometries in the *third* order of t/U expansion[224].) Taking the strong-coupling limit of such a model, one derives[204] a coupled dynamics of interacting spins and *electric dipoles* on the triangular lattice. In turns out that sufficiently strong inter-dimer Coulomb interaction stabilizes charge-ordered state (dipolar solid) and suppresses spin ordering via non-trivial modification of exchange interactions J, J'.

Clearly many more studies, both experimental and theoretical, are required in order to elucidate the physics behind apparent spin-liquid behavior of organic Mott insulators.

In place of conclusion we just state the obvious: despite many years of investigations, quantum magnets on triangular

lattices continue to surprise us. There are no doubts that future studies of new materials and models, inspired by them, will bring out new quantum states and phenomena.

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